ELEC3104: Digital Signal Processing

Topic 3: Digital Filter Fundamentals

- FIR and IIR filters (Ref pp. 111-117)
- Digital Filter Structures (Ref pp. 334-343)
- Magnitude and Phase Responses (Ref pp. 308-352)
- Minimum, Maximum, Mixed Phase Filters (Ref PP. 362-368)
- All-Pass filters (Ref pp. 353-354)

Reference Book: J. Proakis & D. Manolakis, Digital Signal Processing, Prentice-Hall, 2014

Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

Digital Filter Fundamentals

Introduction

- A digital filter is a discrete-time system designed to modify selected frequency components of a signal.
- There are two types of digital filters:
  (a) FIR (Finite Impulse Response) filters - Non-recursive filters (no feedback path)

  (b) IIR (Infinite Impulse Response) filters - Recursive filters (at least one feedback path present)
- A linear time invariant (LTI) discrete-time system described by the following equation is commonly called a digital filter:

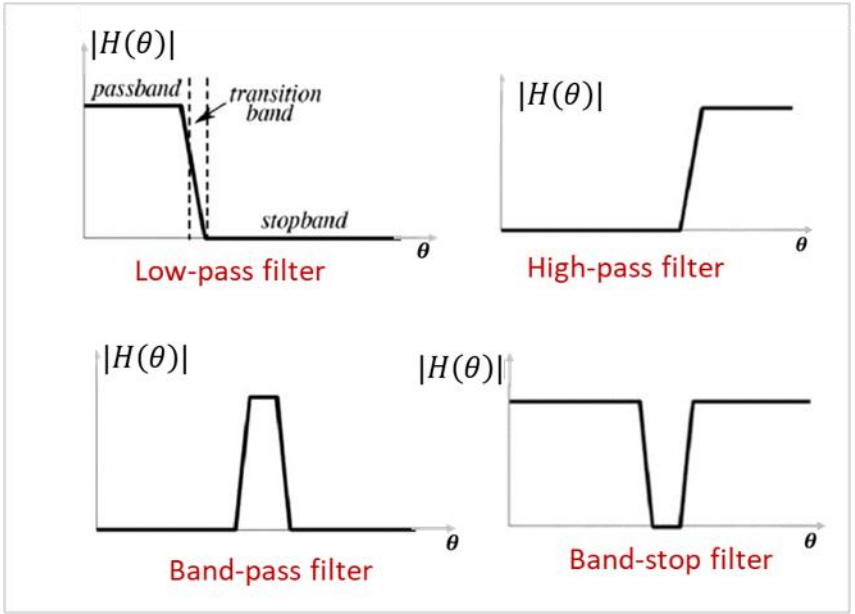
$$y[n] = \sum_{k=0}^{M} a_k x[n-k] \quad - \quad \sum_{k=1}^{L} b_k y[n-k]$$

Feedforward Feedback

where $x[n]$ is the input signal, $y[n]$ is the output signal and $a_0, a_1, a_2, \ldots, a_M$ and $b_1, b_2, b_3, \ldots, b_L$ are constants (filter coefficients). These coefficients determine the characteristics of the filter.

- When all $b_k = 0$, there is no feedback and the filter is FIR. When one or more $b_k \neq 0$, the filter is recursive and is typically an IIR filter.





Four Common Digital Filters

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Digital Filter Fundamentals: FIR filter

FIR filter

- Any filter with an impulse response of finite duration is called a Finite Impulse Response (FIR) filter.
- If all $b_k = 0$, then the calculation of *y[n]* does not require the use of previously calculated samples of the output (see the equation below):

$$y[n] = \sum_{k=0}^{M} a_k x[n-k] \quad - \quad \sum_{k=1}^{L} b_k y[n-k]$$

$$\therefore \quad y[n] \quad = \quad \sum_{k=0}^{M} a_k x[n-k]$$

$$= \quad a_0 x[n] + a_1 x[n-1] + \cdots + a_M x[n-M]$$

- This is recognised as a convolution sum:

$$y[n] = \sum_{k=0}^{M} h[k] x[n-k]$$

- Therefore, the impulse response is identical to the coefficients, that is,

$$h[n] = \begin{cases} a_n & 0 \le n \le M \\ 0 & otherwise \end{cases}$$

Example 1

- The following is a (non-recursive) second order FIR filter:

$$y[n] \quad = a_0 x[n] + a_1 x[n-1] + a_2 x[n-2]$$

$$H(z) = \frac{Y(z)}{X(z)} = a_0 + a_1 z^{-1} + a_2 z^{-2}$$

$$h[n] = \{ a_0, a_1 , a_2 \}$$ ← Impulse response

FIR filter properties

A property of the FIR filter is that it will always be stable:

- **Stability** requires that there should be no poles outside or on the unit circle. This condition is automatically satisfied for an FIR filter. In fact, all its poles are located at the origin.
- **Note:** An FIR filter is always stable because its impulse response has only a finite number of non-zero samples. Therefore $\sum_{n=-\infty}^{n=\infty} |h[n]| < \infty$ which satisfies the condition for BIBO stability.
- **Linear Phase:** An important property of FIR filters is that they can be designed to have an exactly **linear phase** response (Not every FIR filter has linear phase), which is highly desirable in many signal processing applications.

Digital Filter Fundamentals: FIR filter – Phase Delay & Group Delay

- When a signal passes through a filter, it is modified in amplitude and/or phase.
- The nature and extent of the modification of the signal is dependent on the amplitude and phase characteristics of the filter.
- The **phase delay** or **group delay** of the filter provides a useful measure of how the filter modified the phase characteristic of the signal.



Phase Delay

- **Phase delay** is the time delay experienced by a sinusoidal component at a particular frequency as it passes through the filter.

$$Phase_delay(T_p) = -\frac{\emptyset(\theta)}{\theta}$$

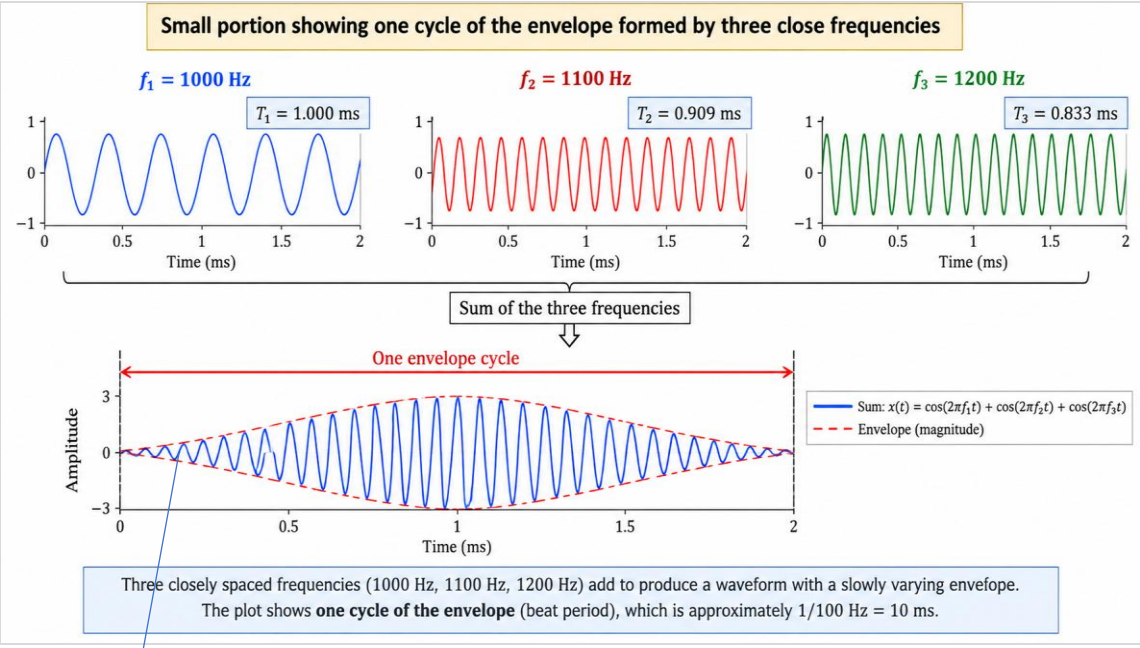
T_p = the negative of the phase angle divided by frequency; where $\phi(\theta)$is the phase response of the filter.

Group Delay

- **Group delay** describes how the time delay introduced by a filter varies with frequency. It is defined as the negative derivative of the filter phase response with respect to frequency:

$$Group_delay(T_g) = -\frac{d\emptyset(\theta)}{d\theta}$$

- For **a narrow group of nearby frequencies**, T_g represents the delay experienced by the envelope of the group.

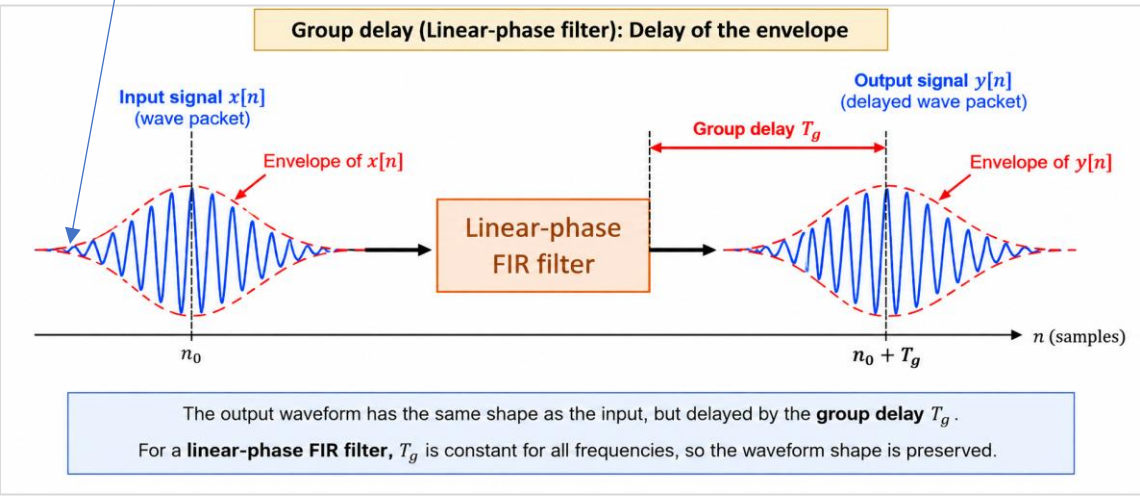


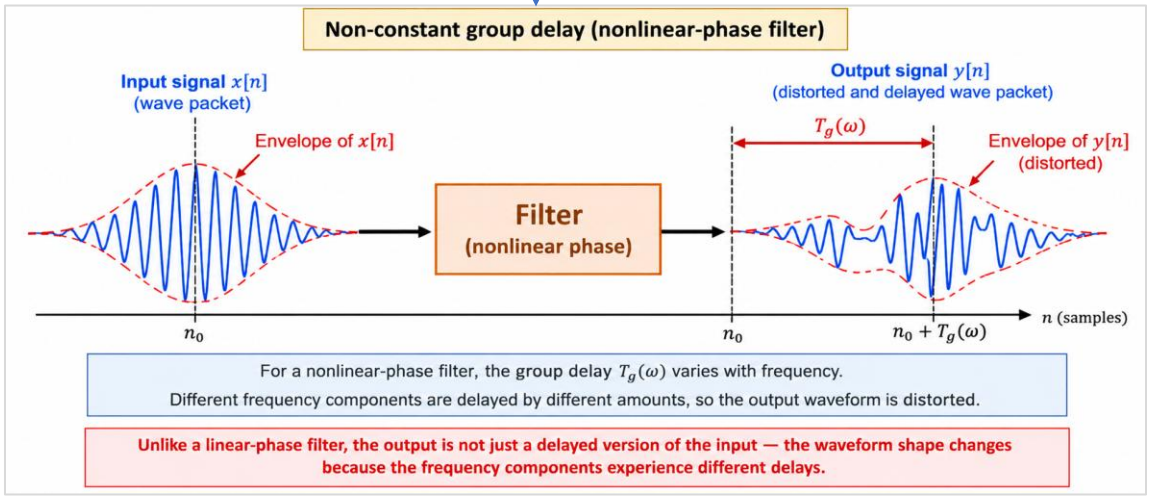
Example

- For example, imagine a musical sound containing components at $500,\ 1000,\ 2000,\ 4000$ Hz.
- With a **linear-phase filter**, they might all effectively experience: $T_g\ (Group\ delay) = 5$ ms.
- The entire musical waveform is simply shifted by 5 ms (apart from the intended magnitude filtering).
- With a nonlinear-phase filter, you might instead have:

$$T_g(500) = 2 \text{ ms}, T_g(1000) = 4 \text{ ms}, T_g(2000) = 7 \text{ ms},$$
$$T_g(4000) = 10 \text{ ms}$$

- Now different spectral regions are delayed differently. When they recombine at the output, their **relative timing/phase relationships have changed**, which can alter transients and waveform shape.





Digital Filter Fundamentals: Linear- Phase FIR filter

Linear Phase Filter

- A *constant group delay* means that signal components at different frequencies experience the same delay in the filter.
- A *linear phase filter* gives same time delay to all frequency components of the input signal. A filter with a nonlinear phase response can cause phase distortion in the output signal.
- This is because different frequency components of the signal experience different time delays, thereby changing their relative phase relationships. Such phase distortion is undesirable in many applications, for example music and video.
- A filter is said to have a **linear phase response** if its phase response satisfies one of the following relationships:

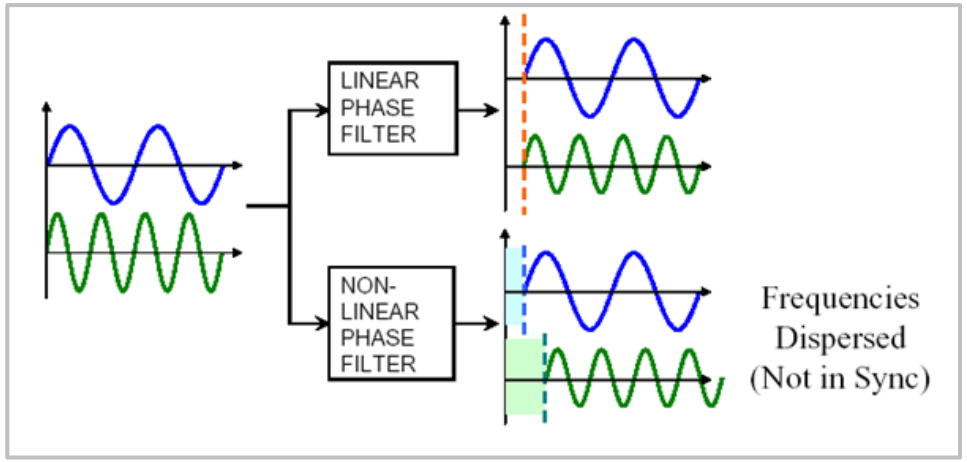
$$\emptyset(\theta) = -a\theta$$

$$\emptyset(\theta) = b - a\theta; \quad \text{a \& b are constants}$$

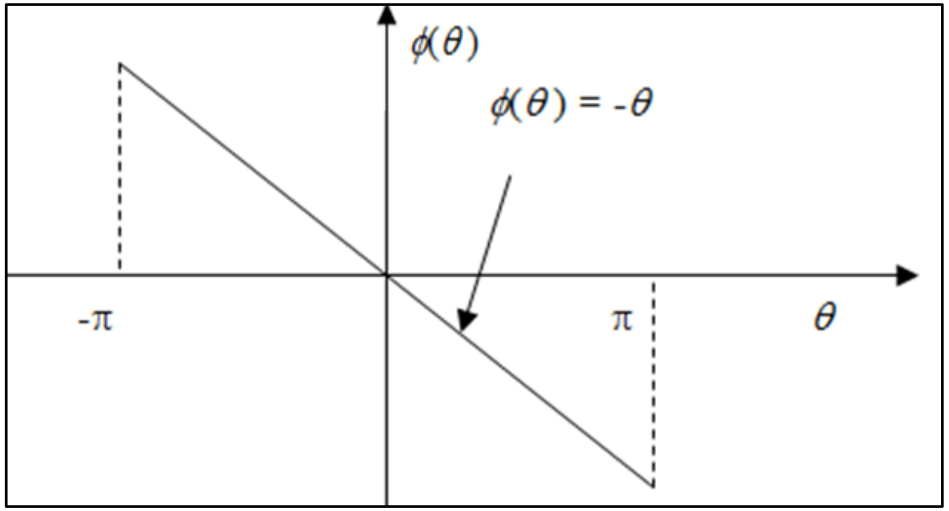
From $T_g = -\frac{d\emptyset(\theta)}{d\theta}$; Both equations above give $T_g = a$;

so the group delay is constant.

Comparing a linear phase filter with a non-linear phase filter



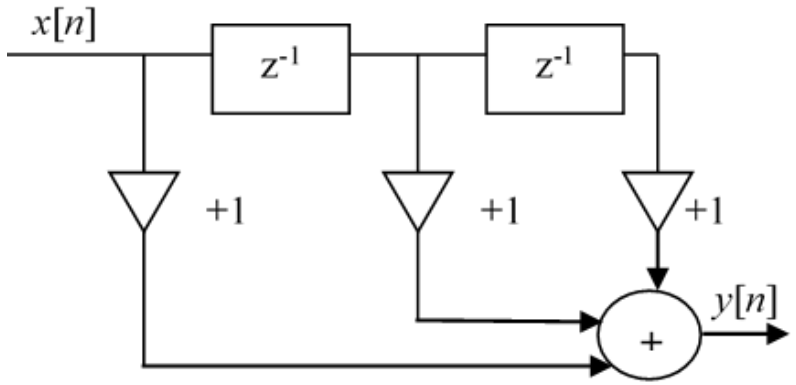
Phase Response of a linear Phase filter



Phase varies linearly with frequency

Digital Filter Fundamentals: Linear- Phase FIR filter

Example 2: Two filter structures are shown below. Show that both filters have linear phase.



$$y[n] = x[n] + x[n-1] + x[n-2]$$

$$H(z) = 1 + z^{-1} + z^{-2}$$

$$H(\theta) = 1 + e^{-j\theta} + e^{-j2\theta}$$

$$= e^{-j\theta}\left(e^{j\theta} + e^{-j\theta} + 1\right)$$

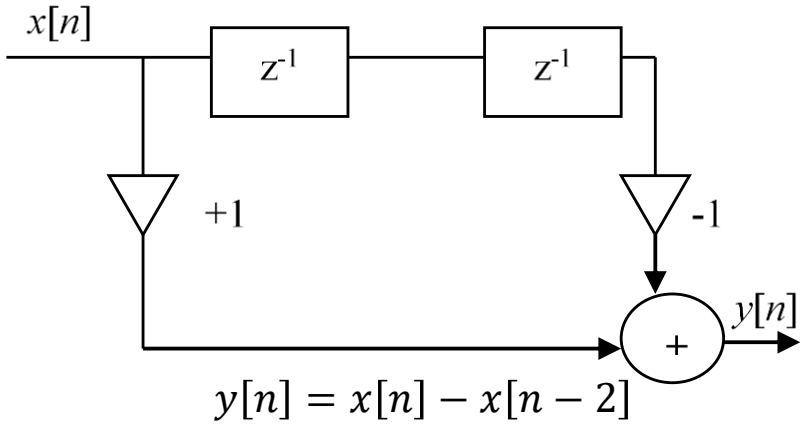
$$= e^{-j\theta}(1 + 2\cos\theta)$$

Let $B(\theta) = 1 + 2cos\theta$

Phase: $\phi(\theta) = -\theta \quad B(\theta) > 0$

$\phi(\theta) = -\theta \ \pm\pi \quad B(\theta) < 0$

- The impulse response $h[n] = [1, 1, 1]$ is even symmetric about its midpoint, resulting in a linear phase response.



$$y[n] = x[n] - x[n-2]$$

$$H(z) = 1 - z^{-2}; \ H(\theta) = 1 - e^{-j2\theta}$$

$$= 2je^{-j\theta}\left(\frac{e^{j\theta} - e^{-j\theta}}{2j}\right) = 2e^{j\frac{\pi}{2}}e^{-j\theta}\sin\theta$$

$$= e^{j\left(\frac{\pi}{2}-\theta\right)}(2\sin\theta)$$

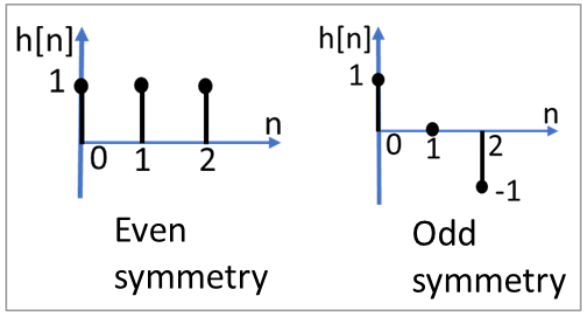
Let $B(\theta) = 2sin\theta$

Phase: $\phi(\theta) = \pi/2 - \theta \quad B(\theta) > 0$

$\phi(\theta) = (\pi/2 - \theta) \pm\pi \quad B(\theta) < 0$

- The impulse response $h[n] = [1, 0, -1]$ is odd symmetric about its midpoint, resulting in a linear phase response.

For a linear phase FIR filter its coefficients are symmetrical about its midpoint



General form for a linear-phase FIR filter

- The frequency response of a linear-phase FIR filter can be written as

$$H(\theta) = A(\theta)e^{j\phi(\theta)}$$

where $A(\theta)$is real and the phase is linear with frequency.

- For an even **symmetric impulse response**

$$H(\theta) = A(\theta)e^{-j\alpha\theta}$$

- For an odd **symmetric impulse response**

$$H(\theta) = jA(\theta)e^{-j\alpha\theta}$$

or equivalently, $H(\theta) = A(\theta)e^{j\left(\frac{\pi}{2}-\alpha\theta\right)}$

where $A(\theta)$is real and α is a constant.

For an FIR filter of order M:

Even Symmetry:

$$\boxed{h[n] = h[M-n]}$$

The coefficients mirror each other with the **same sign**.

For example,

$$h[n] = [1, 2, 3, 2, 1]$$

Odd symmetry:

$$\boxed{h[n] = -h[M-n]}$$

The coefficients mirror each other with **opposite signs**.

For example,

$$h[n] = [1, 2, 0, -2, -1].$$

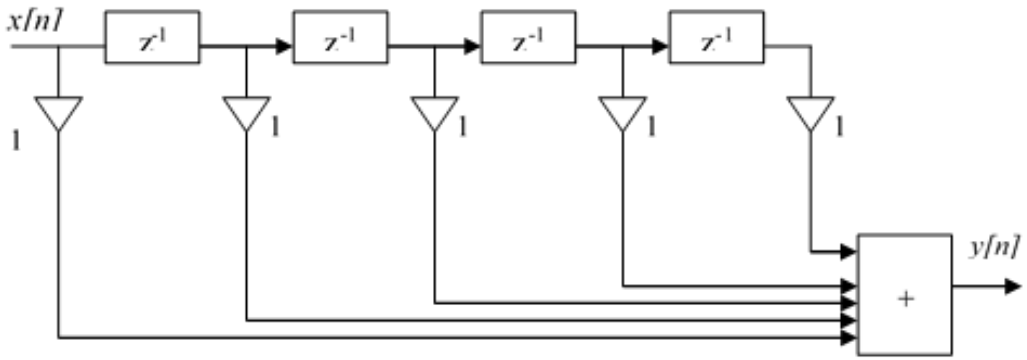
For an FIR filter of order M, Even Symmetry or odd symmetry about its midpoint, $\alpha = \frac{M}{2}$

Digital Filter Fundamentals: Linear phase FIR filter

Example 3:

a) Show that the filter structure shown on the right has a linear phase response given by: $\phi(\theta) = -2\theta$.

b) Show that, for the above linear- phase filter , both the phase delay and the group delay are constant.



Answer:

$$H(z) = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4}$$

Let N = number of coefficients of the liner phase filter

$$H(z) = \frac{1 - z^{-N}}{1 - z^{-1}} \Rightarrow H(\theta) = \frac{1 - e^{-j\theta N}}{1 - e^{-j\theta}}$$

$$H(\theta) = \frac{e^{-j\frac{\theta N}{2}}}{e^{-j\frac{\theta}{2}}} \frac{\left[e^{j\frac{\theta N}{2}} - e^{-j\frac{\theta N}{2}}\right]}{\left[e^{j\frac{\theta}{2}} - e^{-j\frac{\theta}{2}}\right]} = e^{-j\frac{N-1}{2}\theta} \frac{\sin\left(\frac{N\theta}{2}\right)}{\sin\frac{\theta}{2}}$$

$$N = 5 \quad \Rightarrow \quad H(\theta) = e^{-j2\theta}\left\{\frac{\sin\frac{5\theta}{2}}{\sin\frac{\theta}{2}}\right\}$$

B(θ)

$\therefore \phi(\theta) = -2\theta \quad B(\theta) > 0$

$\phi(\theta) = -2\theta \pm \pi \quad B(\theta) < 0$

$$Phase_delay(T_p) = -\frac{\varphi(\theta)}{\theta} = -\frac{-2\theta}{\theta}$$

$$= 2\ (constant)$$

$$Group_delay(T_g) = -\frac{d\varphi(\theta)}{d\theta} = -\frac{d[-2\theta)]}{d\theta}$$

$$= 2\ (constant)$$

Digital Filter Fundamentals: IIR filter

IIR filter

- A recursive digital filter contains at least one feedback path (closed loop), which includes at least one delay element.

$$y[n] = \sum_{k=0}^{M} a_k x[n-k] \quad - \quad \sum_{k=1}^{L} b_k y[n-k]$$

- For a recursive digital filter at least one $b_k \neq 0$. For example, *if* $a_k = 0$ for $k \geq 1$ and only $b_1 \neq 0$, then

$$\therefore \quad y[n] = a_0 x[n] - b_1 y[n-1]$$

$$H(z) = \frac{a_0}{1+b_1 z^{-1}} \qquad \text{[IIR filter]}$$

- A recursive filter is generally an infinite impulse response filter (IIR).

Second Order Filters

Transfer function	Description
$H_1(z) = a_0 + a_1 z^{-1} + a_2 z^{-2}$	all-zero second order FIR filter (excluding poles at origin)
$H_2(z) = \dfrac{1}{1 + b_1 z^{-1} + b_2 z^{-2}}$	all-pole second order IIR filter (excluding zeros at origin)
$H_3(z) = \dfrac{1 + a_1 z^{-1} + a_2 z^{-2}}{1 + b_1 z^{-1} + b_2 z^{-2}}$	pole-zero second order IIR filter

Note: Poles and zeros can be real or complex

Digital Filter Fundamentals: IIR filter – DC Gain and 3 dB Bandwidth

Example 4: DC gain calculation

- Consider the 1st order IIR filter

$$H(z) = \frac{1}{1 - az^{-1}} \quad 0 < a < 1$$

The difference equation is:

$$y[n] = x[n] + a\,y[n-1].$$

- The DC gain of $H(z)$ can be obtained by substituting $\theta = 0$.

$$H(\theta)|_{\theta=0} = \frac{1}{1 - ae^{-j(0)}} = \frac{1}{1-a}$$

- To obtain unity DC gain, introduce a gain factor of $(1-a)$, so that $H(z)$ becomes

$$H(z) = \frac{1-a}{1-az^{-1}} \quad \text{[DC gain =1]}$$

$$\therefore y[n] = (1-a)\,x[n] + a\,y[n-1]$$

Example 5: 3-dB cutoff frequency and bandwidth calculation

- Consider a lowpass filter $y[n] = ay[n-1] + bx[n], 0 < a < 1$

(i) Determine '*b*' so that $|H(0)| = 1$.
(ii) Determine the 3dB bandwidth for the normalised filter in part (i).

Answer:

$$Y(z) = a\,Y(z)z^{-1} + b\,X(z) \Rightarrow H(z) = \frac{b}{1-az^{-1}} \Rightarrow H(\theta) = \frac{b}{1-ae^{-j\theta}}$$

we have $|H(0)| = 1;\ H(0) = \frac{b}{1-a} = 1 \Rightarrow b = 1 - a$

(ii) $H(\theta) = \frac{1-a}{1-ae^{-j\theta}} = \frac{1-a}{(1-a\cos\theta)+ja\sin\theta}; |H(\theta)| = \frac{1-a}{\sqrt{1+a^2-2a\cos\theta}}$

Second Method: $|H(\theta)|^2 = H(\theta)\cdot H^*(\theta)$ {$H^*(\theta)$ is the complex conjugate of $H(\theta)$}

$$\left\{|H(\theta)|^2 = \frac{1-a}{1-ae^{-j\theta}}\cdot\frac{1-a}{1-ae^{j\theta}} = \frac{(1-a)^2}{1-2a\cos\theta+a^2}\right\}$$

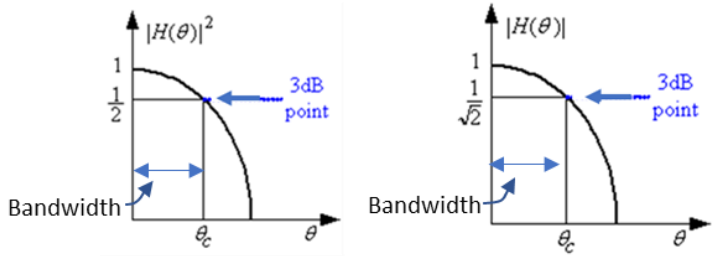
θ_c is the cut-off frequency at which:

$$|H(\theta)|^2_{\theta=\theta_c} = \frac{1}{2}|H(0)|^2 \quad \textbf{or} \quad |H(\theta)|_{\theta=\theta_c} = \frac{1}{\sqrt{2}}|H(0)|$$

We have $|H(0)|^2 = 1$ *when* $\theta = 0$;

$$\frac{1}{2} = \frac{(1-a)^2}{1+a^2-2a\cos\theta_c}$$

$$\theta_c = \cos^{-1}\left[\frac{-a^2+4a-1}{2a}\right]$$



Digital Filter Fundamentals: Digital filter Structures

Digital Filter Structures

- A digital filter can be implemented using different structures, such as Direct Form I and Direct Form II. More complex filters can also be realised by combining simpler filter sections.
- There are two common ways of combining simple filter sections, either in **cascade** or in **parallel**.

$$y[n] = \sum_{k=0}^{M} a_k x[n-k] - \sum_{k=1}^{L} b_k y[n-k]$$

$$Y(z) = \sum_{k=0}^{M} a_k X(z) z^{-k} - \sum_{k=1}^{L} b_k Y(z) z^{-k}$$

$$\therefore H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} a_k z^{-k}}{1 + \sum_{k=1}^{L} b_k z^{-k}}$$

- **Direct form I:**

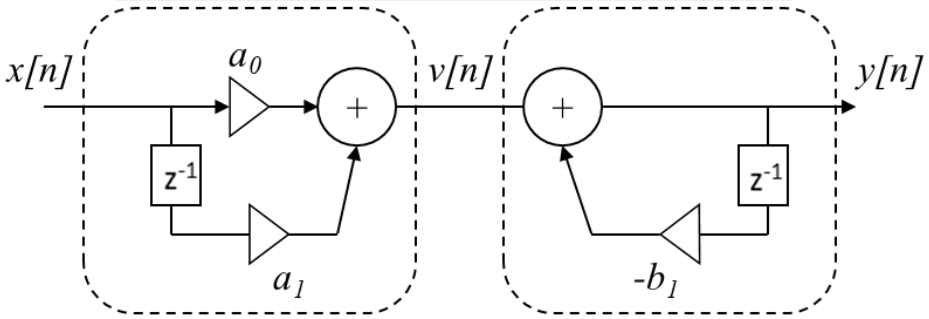
$$H(z) = \underbrace{\sum_{k=0}^{M} a_k z^{-k}}_{zeros} \cdot \underbrace{\frac{1}{1 + \sum_{k=1}^{L} b_k z^{-k}}}_{poles}$$

- **Alternative Structure for Direct form I:**

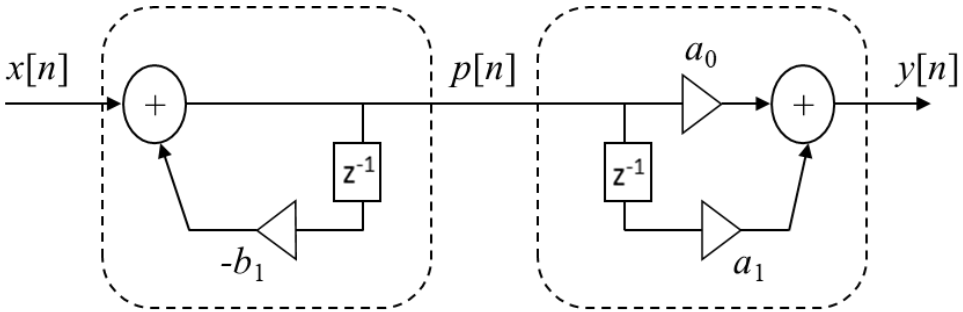
$$H(z) = \underbrace{\frac{1}{1 + \sum_{k=1}^{L} b_k z^{-k}}}_{poles} \cdot \underbrace{\sum_{k=0}^{M} a_k z^{-k}}_{zeros}$$

Direct form I

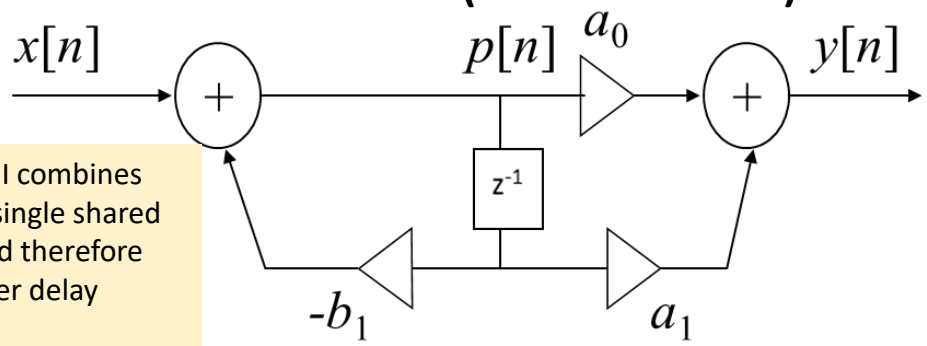
Direct Form I uses separate delay lines for the feedforward and feedback sections.



Alternative Structure for Direct form I



Direct form II (Canonical Form)



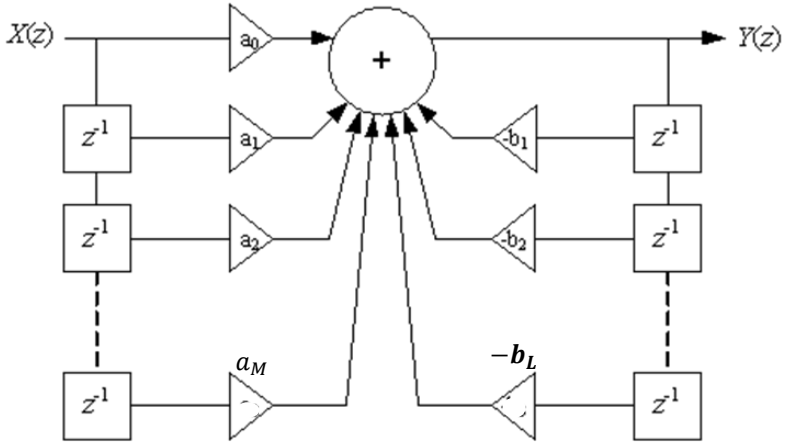
Direct Form II combines them into a single shared delay line and therefore requires fewer delay elements.

Digital Filter Fundamentals: Digital filter Structures

Direct Form I:

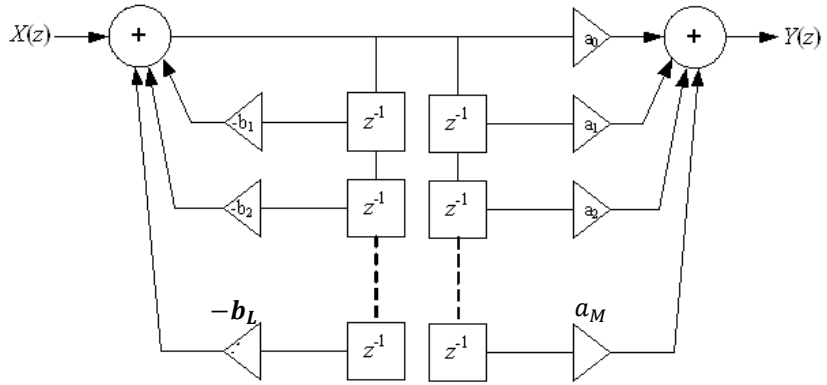
- **Direct form I:**

$$H(z) = \underbrace{\sum_{k=0}^{M} a_k z^{-k}}_{zeros} \cdot \underbrace{\frac{1}{1+\sum_{k=1}^{L} b_k z^{-k}}}_{poles}$$



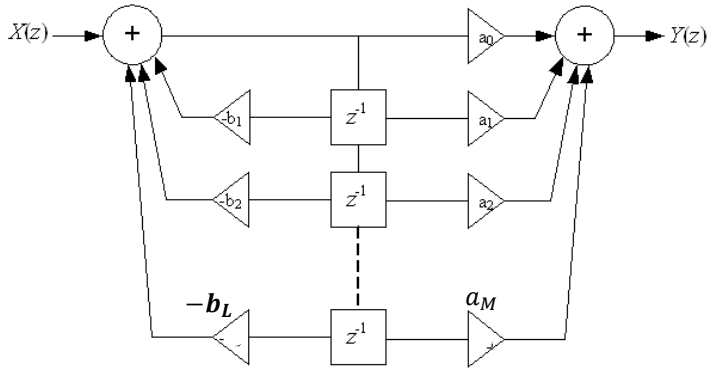
Alternative Structure for Direct form I

$$H(z) = \underbrace{\frac{1}{1+\sum_{k=1}^{L} b_k z^{-k}}}_{poles} \cdot \underbrace{\sum_{k=0}^{M} a_k z^{-k}}_{zeros}$$



Direct Form II (Canonic Realisation)

- Direct Form II is a canonic realisation because it uses the minimum number of delay elements. For $M = L$, only M delay elements are required as shown below.

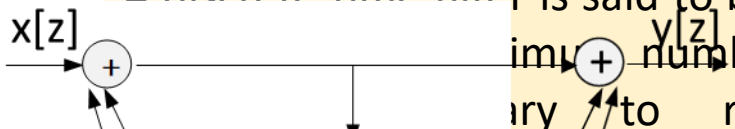


Direct Form I has two separate delay lines:

- M delays for the feedforward section
- L delays for the feedback section

so it requires $M + L$ delay elements.

A discrete-time filter is said to be canonic if imu number of delay ary to realise the

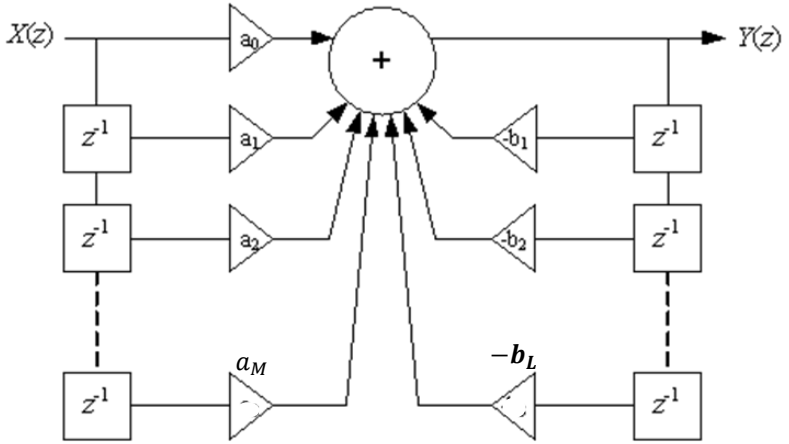


Digital Filter Fundamentals: Digital filter Structures

Direct Form I:

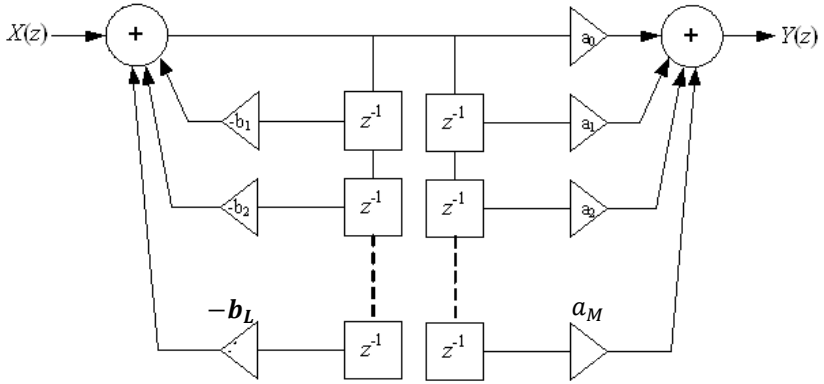
- **Direct form I:**

$$H(z) = \underbrace{\sum_{k=0}^{M} a_k z^{-k}}_{zeros} \cdot \underbrace{\frac{1}{1+\sum_{k=1}^{L} b_k z^{-k}}}_{poles}$$



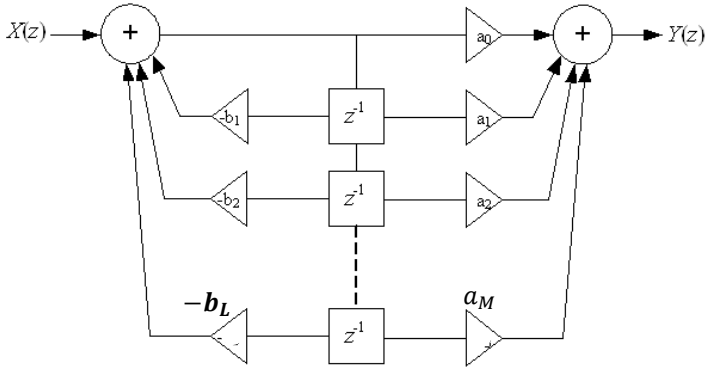
Alternative Structure for Direct form I

$$H(z) = \underbrace{\frac{1}{1+\sum_{k=1}^{L} b_k z^{-k}}}_{poles} \cdot \underbrace{\sum_{k=0}^{M} a_k z^{-k}}_{zeros}$$



Direct Form II - Canonic Realisation

- Direct Form II is a canonic realisation because it uses the minimum number of delay elements. For $M = L$, only M delay elements are required as shown below.



Direct Form I has two separate delay lines:

- M delays for the feedforward section
- L delays for the feedback section

so it requires $M + L$ delay elements.

Direct Form II rearranges the structure so that the same delay line is shared by the feedforward and feedback parts. It therefore requires only

$$\boxed{max(M, L)}$$

delay elements. That is the minimum number needed, so Direct Form II is a canonic realisation.

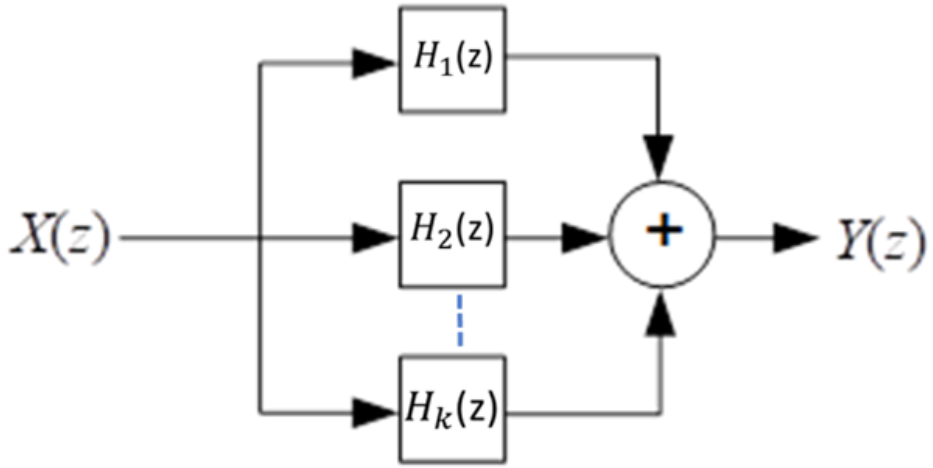
Digital Filter Fundamentals: IIR filter – Parallel Structure

Parallel Structure

$$H(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} + \cdots + a_M z^{-M}}{1 + b_1 z^{-1} + b_2 z^{-2} + \cdots + b_L z^{-L}}$$

$$= \underbrace{\sum_{i=1}^{k} H_i(z) \quad = \quad H_1(z) + H_2(z) + H_3(z) + \ldots + H_k(z)}_{parallel\ structure}$$

Note: Use partial fraction decomposition to obtain $H_i(z)$



Parallel Structure

Example 7:

- Draw a parallel realisation of a third order system $H(z)$ given below: $H(z) = \frac{z^3 - z^2 - 5z - 3}{5z(z^2 - 4)}$

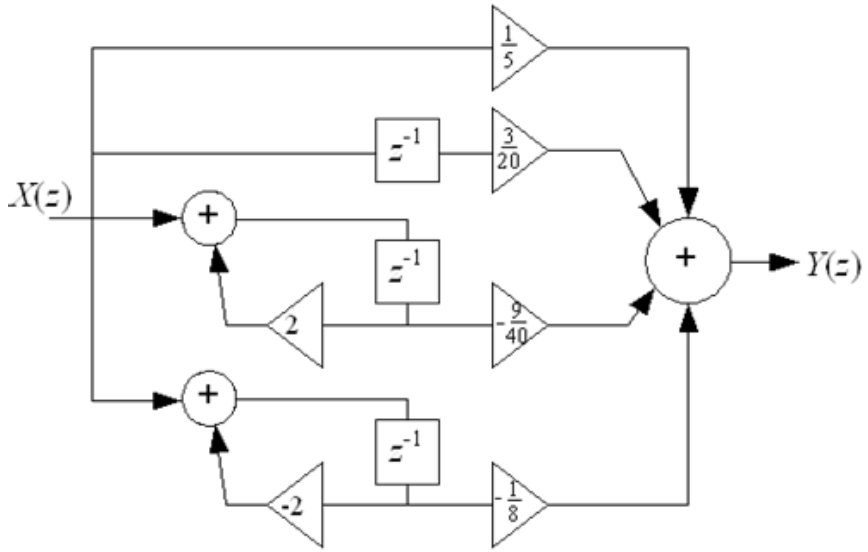
Answer:

By using a partial fraction decomposition, $H(z)$ becomes:

$$H(z) = A + \frac{B}{z} + \frac{C}{z-2} + \frac{D}{z+2}$$

$$H(z) = \frac{1}{5} + \frac{3}{20z} - \frac{9}{40(z-2)} - \frac{1}{8(z+2)}$$

$$H(z) = \frac{1}{5} + \frac{3}{20} z^{-1} - \frac{9z^{-1}}{40(1-2z^{-1})} - \frac{z^{-1}}{8(1+2z^{-1})}$$



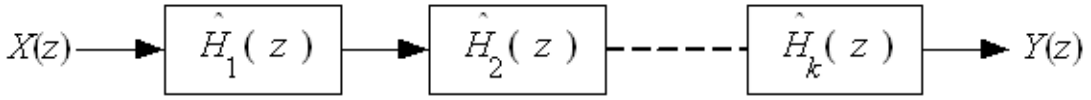
Digital Filter Fundamentals: IIR filter – Cascade Structure

Cascade Structure

$$H(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} + \cdots + a_M z^{-M}}{1 + b_1 z^{-1} + b_2 z^{-2} + \cdots + b_L z^{-L}}$$

$$H(z) = \prod_{i=1}^{k} \hat{H}_i(z) = \underbrace{\hat{H}_1(z) \cdot \hat{H}_2(z) \cdot \hat{H}_3(z) \ldots \hat{H}_k(z)}_{cascade\ structure}$$

Note: H(z) is factored into a product of lower-order transfer functions $\hat{H}_1(z), \hat{H}_2(z), \ldots$, typically first- or second-order sections. These sections are then connected in cascade.The cascade structure is the most popular form in digital filter implementation.



Cascade Structure

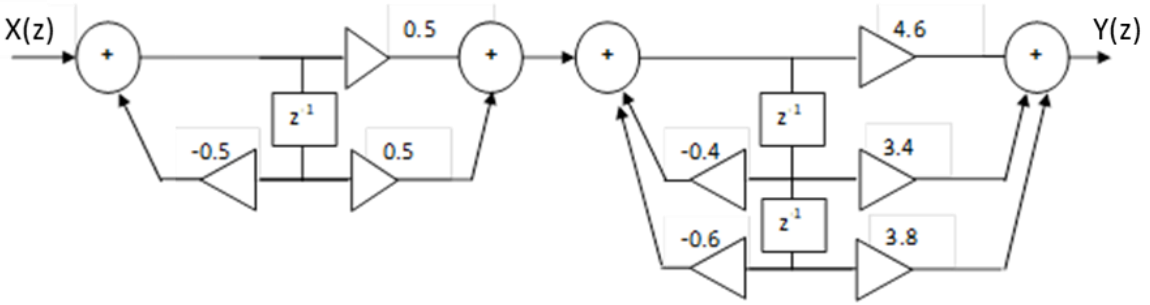
Example 8:

- Obtain a cascade realisation of the following third-order filter:

$$H(z) = \frac{23 + 40z^{-1} + 36z^{-2} + 19z^{-3}}{10 + 9z^{-1} + 8z^{-2} + 3z^{-3}} = \frac{(1 + z^{-1})}{(2 + z^{-1})} \frac{23 + 17z^{-1} + 19z^{-2}}{5 + 2z^{-1} + 3z^{-2}}$$

$$= \frac{(1 + z^{-1})}{2(1 + 0.5z^{-1})} \frac{23 + 17z^{-1} + 19z^{-2}}{5(1 + 0.4z^{-1} + 0.6z^{-2})} = \left(\frac{(0.5 + 0.5z^{-1})}{1 + 0.5z^{-1}}\right)\left(\frac{4.6 + 3.4z^{-1} + 3.8z^{-2}}{1 + 0.4z^{-1} + 0.6z^{-2}}\right)$$

1st order 2nd order



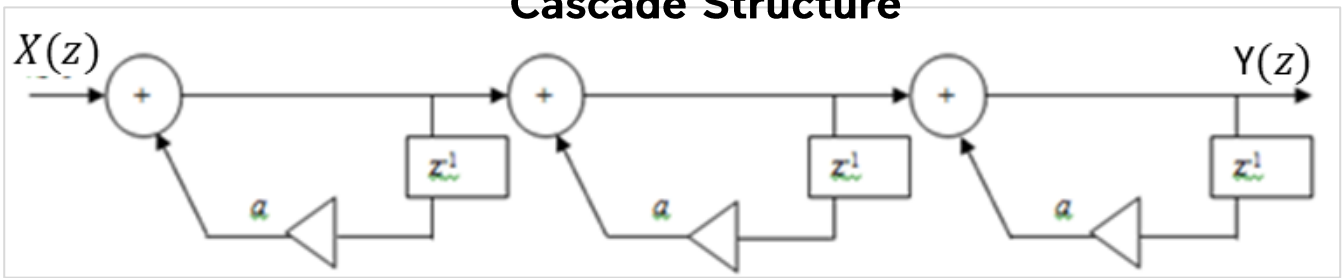
Example 9:

- Draw a cascade realisation of the filter H(z):

$$H(z) = \frac{1}{(1 - az^{-1})^3}$$

Note: The filter can be realised as three identical first-order sections connected in cascade.

Cascade Structure



ELEC3104: Digital Signal Processing

Topic 3: Digital Filter Fundamentals

- FIR and IIR filters (Ref pp. 111-117)
- Digital Filter Structures (Ref pp. 334-343)
- Magnitude and Phase Responses (Ref pp. 308-352)
- Minimum, Maximum, Mixed Phase Filters (Ref PP. 362-368)
- All-Pass filters (Ref pp. 353-354)

Reference Book: J. Proakis & D. Manolakis, Digital Signal Processing, Prentice-Hall, 2014

Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

Digital Filter Fundamentals: Magnitude and Phase Responses

Phase Response

- We can show that the magnitude response is an **even** function of frequency. {| $H(-\theta)$ |=| $H(\theta)$ |}
- The phase response is an **odd** function of frequency. $\{\phi(-\theta) = -\phi(\theta)\}$
- For example, let us consider the following linear-phase form of the transfer function:

$$H(\theta) = e^{-jk\theta} B(\theta) \quad k > 0$$

B(θ) is a real-valued function that can take positive or negative values.

$$H(\theta) = B(\theta)\, cos(-k\theta) + B(\theta) j\, sin(-k\theta)$$

$$H(\theta) = B(\theta)\, cos(k\theta) - jB(\theta)\, sin(k\theta)$$

Let the phase angle be ϕ:

$$tan\,\phi = -\frac{B(\theta)\, sin(k\theta)}{B(\theta)\, cos(k\theta)} = -\,tan(k\theta)$$

$\therefore$ $\tan\phi = \tan(-k\theta) \Rightarrow$ ***ϕ(θ) = -kθ + mπ*** *(m– integer)*

- The phase function $\phi(\theta)$ includes linear phase term and also accommodates for the sign changes in $B(\theta)$.
- Since -1 can be expressed as $e^{\pm j\pi}$, phase jumps of ±π will occur at frequencies where $B(\theta)$ changes sign. Therefore,

If $B(\theta) > 0$, then $\phi(\theta) = -k\theta$.

If $B(\theta) < 0$, then $\phi(\theta) = -k\theta \pm \pi$.

At $B(\theta) = 0, H(\theta) = 0$, so **the phase is undefined**.

Phase Jumps

- A jump of $\pm 2\pi$ may occur when the phase is wrapped to remain within the principal-value range[-π and π]
- A phase change of ± π occurs when $B(\theta)$ changes sign.
- The sign of the phase jump is chosen such that the resulting phase function is odd and, after the jump, lies in the range [-π and π]

Digital Filter Fundamentals: Magnitude and Phase Responses

Example 10: Low-pass FIR filter

- Calculate the magnitude and phase responses of the 3-sample averager given by;

$$h[n] = \begin{cases} \frac{1}{3} & -1 \le n \le 1 \\ 0 & otherwise \end{cases}$$

Answer:

$$H(z) = \sum_{n=-\infty}^{\infty} h(n)z^{-n} = \sum_{n=-1}^{1} h(n)z^{-n} = \frac{1}{3}z^{-1} + \frac{1}{3}z^{0} + \frac{1}{3}z^{1}$$

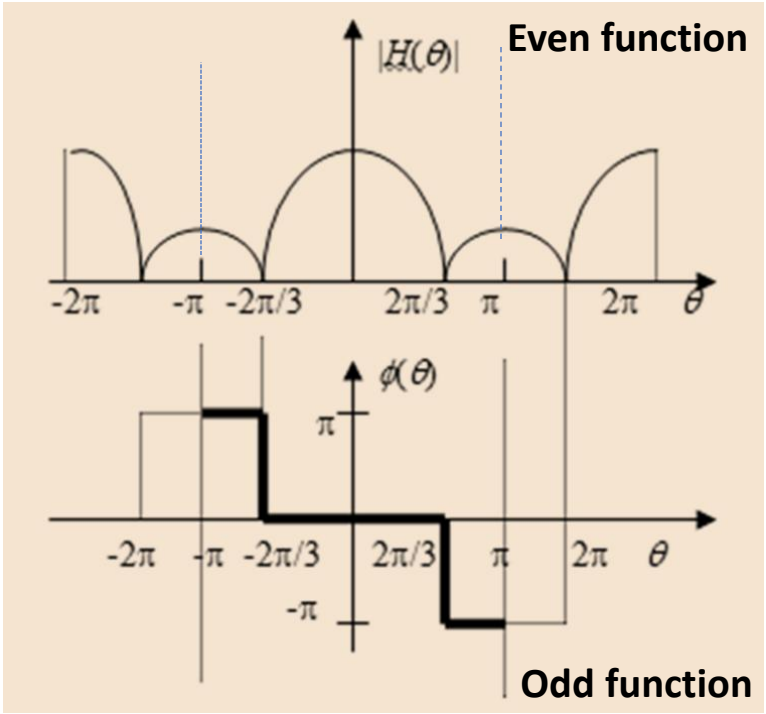
$$H(z) = \frac{1}{3}[z^{-1} + 1 + z^{1}]$$

$$H(\theta) = H(z)\Big|_{z=e^{j\theta}} = \frac{1}{3}\left[e^{-j\theta} + 1 + e^{j\theta}\right] = \left[1 + e^{j\theta} + e^{-j\theta}\right] \cdot \frac{1}{3}$$

$$H(\theta) = \frac{1}{3}[1 + 2\cos\theta] \Rightarrow |H(\theta)| = \left|\frac{1}{3}[1 + 2\cos\theta]\right|$$

$$\begin{cases} \emptyset(\theta) = 0 & H(\theta) \ge 0 \quad -\frac{2\pi}{3} < \theta < \frac{2\pi}{3} \\ \emptyset(\theta) = 0 \pm \pi & H(\theta) < 0 \quad -\pi \le \theta \le -\frac{2\pi}{3} \ \& \ \frac{2\pi}{3} < \theta \le \pi \end{cases}$$

- The appropriate sign of π must be chosen to make $\phi(\theta)$ an odd function of frequency.



- For **odd symmetry** of the phase response:

$$\phi(-\theta) = -\phi(\theta).$$

- At $\theta = \pm\frac{2\pi}{3}$, $H(\theta) = 0$, so the phase is undefined.

Digital Filter Fundamentals: Magnitude and Phase Responses

Example 11: Low-pass FIR filter

- Calculate the magnitude and phase responses of the 3-sample averager given by;

$$h[n] = \begin{cases} \frac{1}{3} & 0 \le n \le 2 \\ 0 & \text{otherwise} \end{cases}$$

Answer:

$$H(z) = \frac{1}{3} + \frac{1}{3}z^{-1} + \frac{1}{3}z^{-2} \Rightarrow H(\theta) = \frac{1}{3} + \frac{1}{3}e^{-j\theta} + \frac{1}{3}e^{-j2\theta}$$

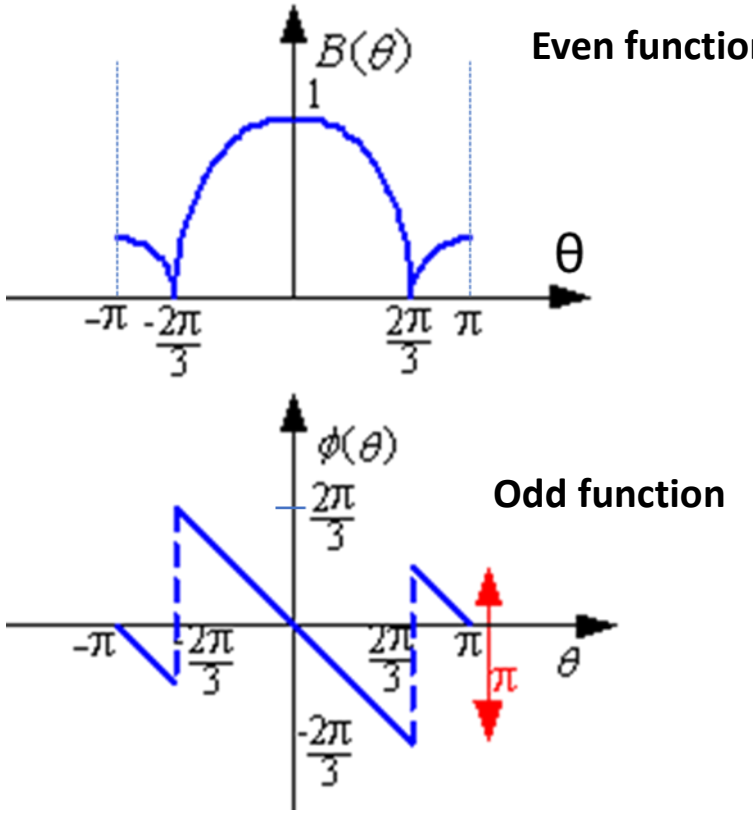
$$H(\theta) = \frac{1}{3}e^{-j\theta}\left[e^{j\theta} + 1 + e^{-j\theta}\right]$$

$$\therefore H(\theta) = e^{-j\theta}\underbrace{\frac{1}{3}[1 + 2\cos\theta]}_{B(\theta)}$$

$$|H(\theta)| = |B(\theta)| = \left|\frac{1}{3}[1 + 2\cos\theta]\right| \quad \text{and} \quad \emptyset(\theta) = -\theta;$$

$$\begin{cases} \emptyset(\theta) = -\theta; & B(\theta) > 0 \quad -\frac{2\pi}{3} < \theta < \frac{2\pi}{3} \\ \emptyset(\theta) = -\theta \pm \pi & B(\theta) < 0 \quad -\pi \le \theta < -\frac{2\pi}{3} \ \& \ \frac{2\pi}{3} < \theta \le \pi \end{cases}$$

- Phase is undefined at points $|H(\theta)| = 0$ or B(θ)=0.



Phase Jumps: A jump of ± π occurs when $B(\theta)$ undergoes a change of sign.

Digital Filter Fundamentals: Magnitude and Phase Responses

Example 12:

- Find the magnitude and phase response of the following:

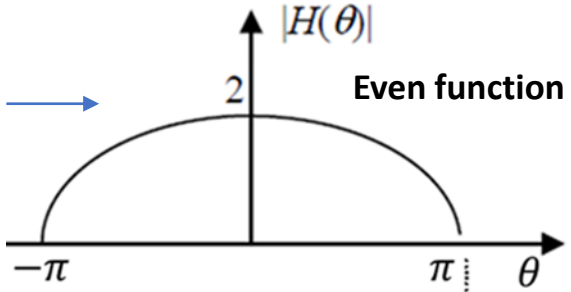
$h(0) = \frac{1}{2}, \quad h(1) = 1, \quad h(2) = \frac{1}{2}, \quad h(n) = 0$, otherwise.

Answer:

$$H(z) = \frac{1}{2}z^0 + 1z^{-1} + \frac{1}{2}z^{-2} = \frac{1}{2} + z^{-1} + \frac{1}{2}z^{-2}$$

$$H(\theta) = \frac{1}{2} + e^{-j\theta} + \frac{1}{2}e^{-j2\theta} = e^{-j\theta}\left[\frac{1}{2}e^{j\theta} + 1 + \frac{1}{2}e^{-j\theta}\right]$$

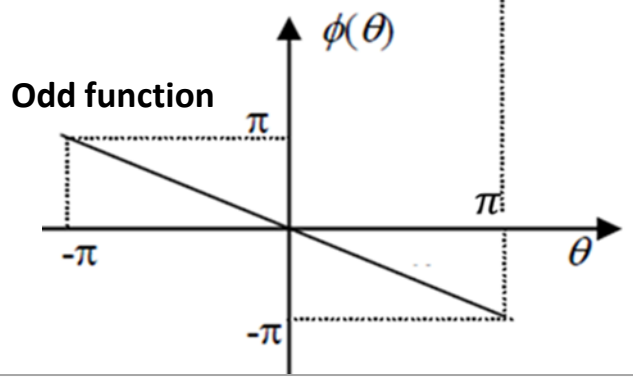
$$H(\theta) = e^{-j\theta}\underbrace{[1 + cos\,\theta]}_{B(\theta)};$$



B(θ)is never negative; therefore, there is no phase jumps of $\pm\pi$.

Phase Response:

$\emptyset(\theta) = -\theta$ $-\pi<\theta<\pi$.

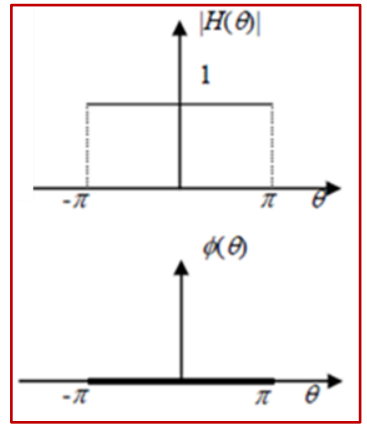


Example 13:

- **Case 1:**

$$h[n] = \delta(n) = 1 \quad n = 0$$
$$h[n] = \delta(n) = 0 \quad otherwise$$

$$\therefore H(\theta) = 1;$$
$$\emptyset(\theta) = 0$$



- **Case 2:** $h[n] = \delta[n-k];$

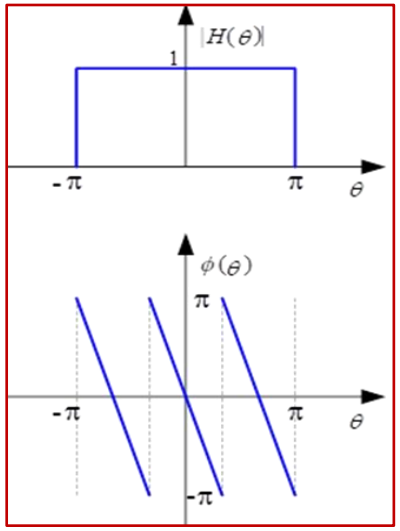
$$\therefore H[z] = 1\,z^{-k};$$

$$H(\theta) = 1\,e^{-jk\theta}$$

$$|H(\theta)| = 1;$$

$$\phi(\theta) = -k\theta;$$

For $k = 3;\ \emptyset(\theta) - 3\theta$



When the phase exceeds the principal range$[-\pi, \pi]$ a phase wrap of $\pm 2\pi$ is used to bring it back into this range.

Digital Filter Fundamentals: Magnitude and Phase Responses

Example 14: Magnitude and Phase Responses of a Low-pass IIR filter

- Determine and sketch the magnitude and phase response of 1st order recursive filter (IIR filter):

$$y[n] = x[n] + ay[n-1] \qquad 0 < a < 1$$

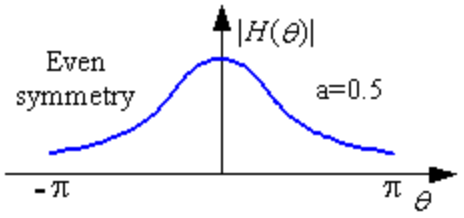
Answer:

$$\frac{Y(z)}{X(z)} = H(z) = \frac{1}{1-az^{-1}}; \qquad H(\theta) = \frac{1}{1-ae^{-j\theta}}$$

$$H(\theta) = \frac{1}{1-ae^{-j\theta}} \cdot \frac{1-ae^{j\theta}}{1-ae^{j\theta}} = \frac{1-a\cos\theta}{1-2a\cos\theta+a^2} - \frac{ja\sin\theta}{1-2a\cos\theta+a^2}$$

$$\tan\emptyset = \frac{-\frac{a\sin\theta}{1-2a\cos\theta+a^2}}{\frac{1-a\cos\theta}{1-2a\cos\theta+a^2}} = \frac{-a\sin\theta}{1-a\cos\theta}$$

Phase: $\emptyset(\theta) = \tan^{-1}\left[\frac{-a\sin\theta}{1-a\cos\theta}\right]$;

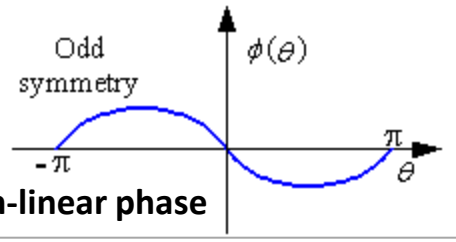


Magnitude:

$$|H(\theta)|^2 = H(\theta).H^*(\theta);$$

$$= \frac{1}{1-ae^{-j\theta}} \cdot \frac{1}{1-ae^{j\theta}}$$

$$|H(\theta)| = \frac{1}{\sqrt{1-2a\cos\theta+a^2}}$$



Non-linear phase

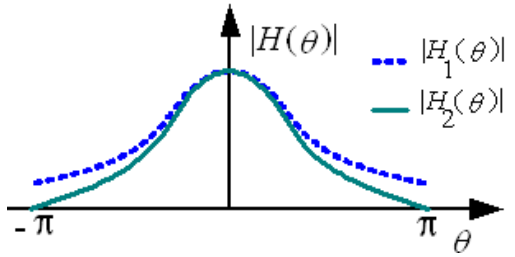
Example 15: Magnitude Responses of IIR filters

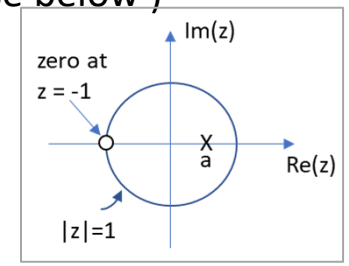
(a) $H_1(z) = \frac{k_0}{1-az^{-1}} \Rightarrow \frac{1-a}{1-az^{-1}} \qquad 0 < a < 1$

The gain k_0 can be selected as $1-a$, so that the **low-pass filter** has unity gain at $\theta = 0$.

(b) $H_2(z) = k_0\frac{1+z^{-1}}{1-az^{-1}}$; In this case, $k_0 = \frac{1-a}{2}$ (for unity gain at $\theta = 0$).

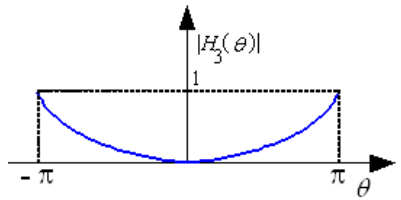
The addition of a zero at $z = -1$ provides zero gain at $\theta = \pi$, thereby increasing the attenuation at high frequencies. (see magnitude response below)

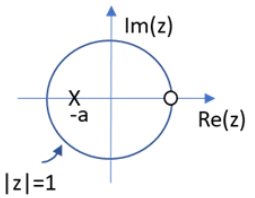




(c) A high-pass filter can be obtained by reflecting the pole-zero locations of the low-pass filter about the imaginary axis in the z-plane.

$$H_3(z) = \frac{1-a}{2} \cdot \frac{1-z^{-1}}{1+az^{-1}}$$





Digital Filter Fundamentals: Magnitude Responses

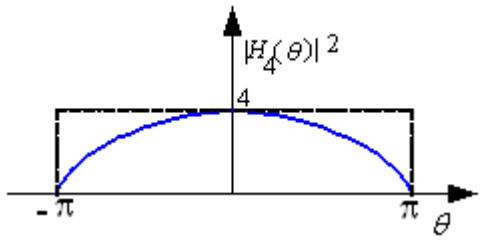
Example 16: FIR filters

- **Low-pass filter:**

$H_4(z) = 1 + z^{-1};$

$y[n] = x[n] + x[n-1]$

$|H_4(\theta)|^2 = 2(1+\cos\theta)$



The two adjacent input samples are added. This reinforces slowly varying (low-frequency) components, while the response becomes zero at $\theta = \pi$.

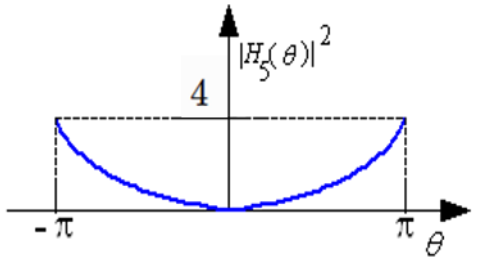
$$|H_4(0)|^2 = 4, \qquad |H_4(\pi)|^2 = 0$$

- **High-pass filter:**

$H_5(z) = 1 - z^{-1};$

$y[n] = x[n] - x[n-1]$

$|H_5(\theta)|^2 = 2(1-\cos\theta)$

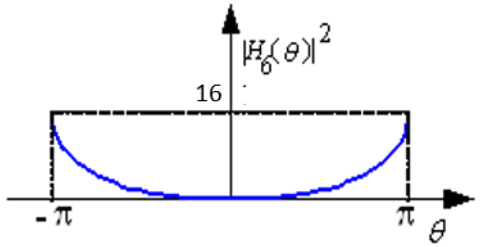


The difference between two adjacent samples is taken. Slowly varying components are suppressed, while high-frequency components are emphasised.

$$|H_5(0)|^2 = 0, \qquad |H_5(\pi)|^2 = 4$$

- **High-pass filter:**

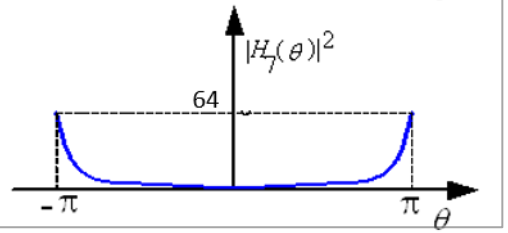
$H_6(z) = (1 - z^{-1})^2$



Repeating the difference operation strengthens the high-pass characteristic. The response becomes flatter near $\theta = 0$and rises more sharply towards high frequencies.

- **High-pass filter:**

$H_7(z) = (1 - z^{-1})^3$



Increasing the order further provides stronger suppression of low frequencies and a sharper high-pass response.

As the number of repeated zeros at $z = 1$increases, low-frequency components are increasingly suppressed and the high-pass characteristic becomes more pronounced.

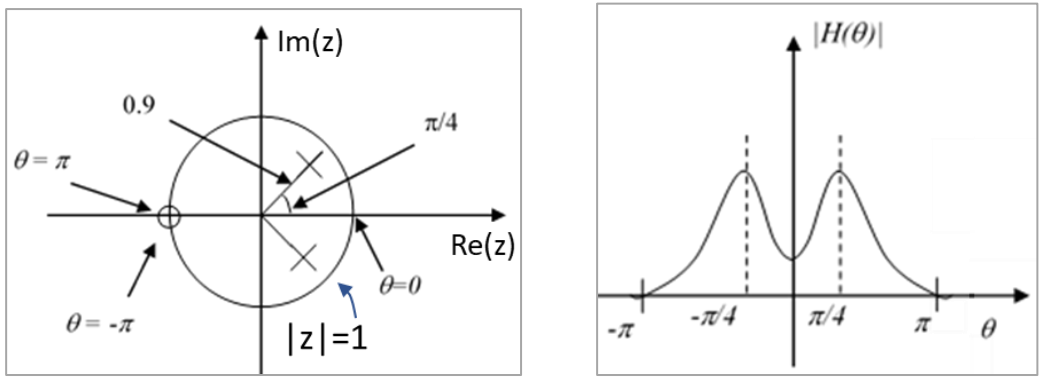
Digital Filter Fundamentals: Magnitude Responses

Example 17: IIR Resonant filter

- Sketch an approximate magnitude response for the system having the transfer function

$$H(z) = \frac{1 + z^{-1}}{(1 - 0.9e^{j\frac{\pi}{4}}z^{-1})(1 - 0.9e^{-j\frac{\pi}{4}}z^{-1})}$$

- The system has a zero at z = -1 & a complex conjugate pair of poles at $z = 0.9\ e^{\pm j\frac{\pi}{4}}$
- Magnitude response will be zero at $\theta = \pi$ and large at $\theta_0 = \pm\ \pi/4$ because the poles are close to the unit circle.



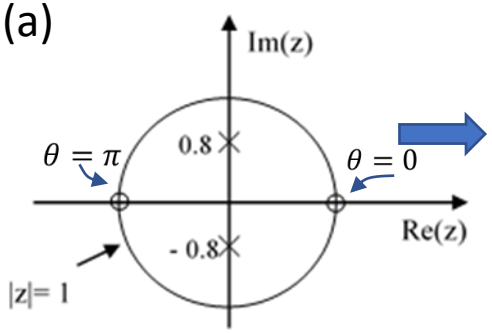
- A complex-conjugate pole pair close to the unit circle produces peaks in the magnitude response near the corresponding pole angles.
- A zero on the unit circle forces the magnitude response $|H(\theta)|$ to zero at the corresponding frequency.

Digital Filter Fundamentals: Magnitude Responses

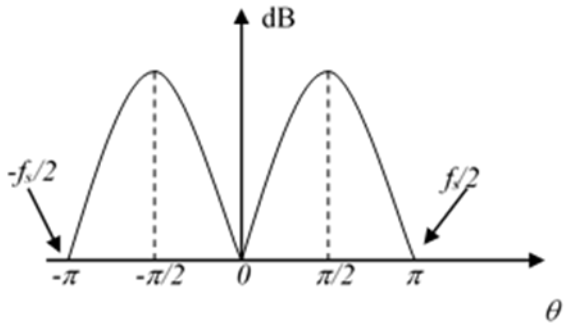
Example 18: IIR resonant filters

- Sketch an approximate magnitude response for each of the pole-zero maps given below:

(a)



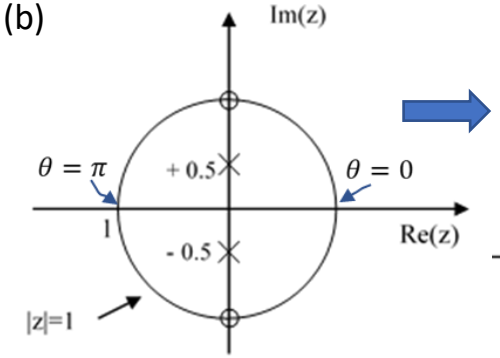
Pole-zero map



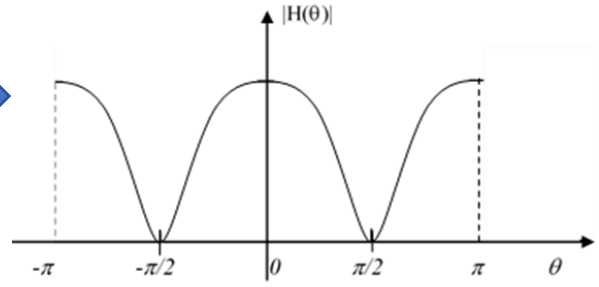
Magnitude response

A complex pole at $\theta_0 = \pm\frac{\pi}{2}$ cause peaks in the magnitude response at $\pm\theta_0$. Zeros on the unit circle at $\theta = 0$ and $\theta = \pi$ will force the magnitude response at those frequencies to zero.

(b)



Pole-zero map



Magnitude response

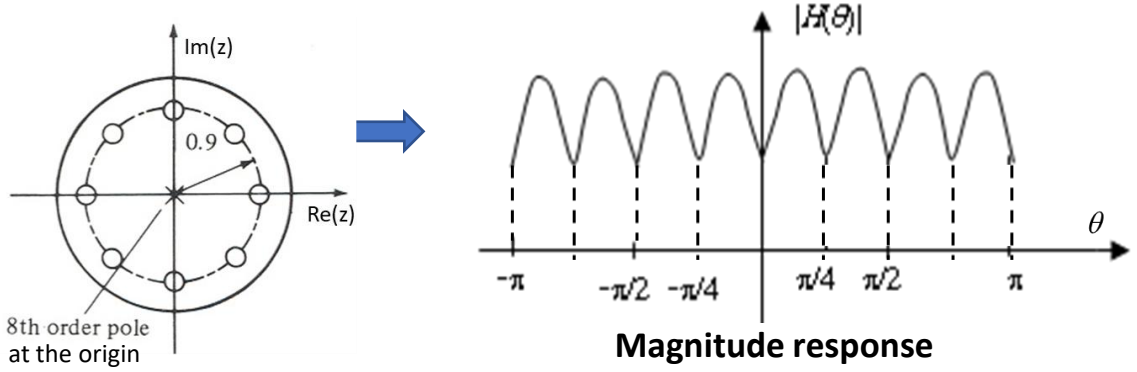
(a) The poles are at approximately $z = \pm j0.8$, so the pole angles are $\theta = \pm\frac{\pi}{2}$, giving resonant peaks near $\pm\pi/2$. The zeros at $z = \pm 1$ force $|H(0)| = 0, |H(\pi)| = 0$.

(b) The poles at approximately $\pm j0.5$ give enhancement around $\pm\pi/2$, while the zeros at $z = \pm j$ force exact zeros at $\theta = \pm\frac{\pi}{2}$.

Digital Filter Fundamentals: Magnitude Responses

Example 19: FIR Magnitude Responses from Pole-Zero Plots

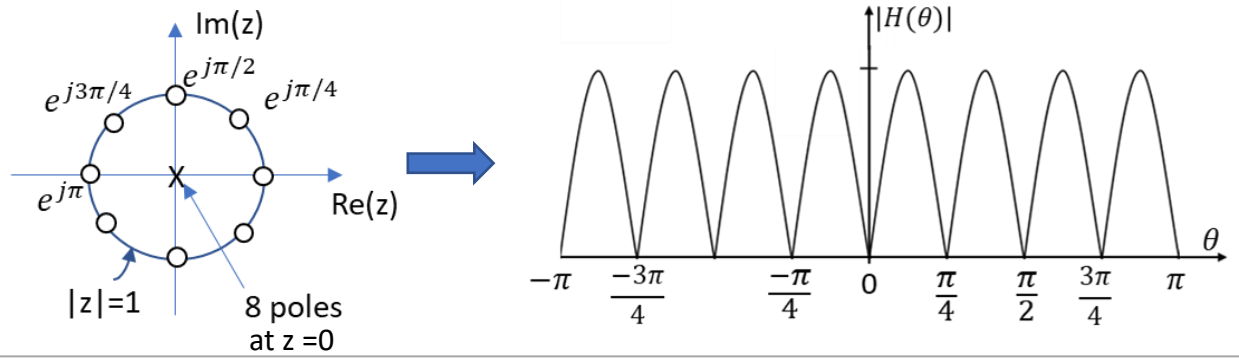
- Sketch roughly the magnitude response corresponding to the pole-zero pattern given below:



Magnitude response

Note: Zeroes near the unit circle cause a dip in the magnitude response.

- Sketch roughly the magnitude response corresponding to the pole-zero pattern shown below, where $N = 8$ zeros are equally spaced around the unit circle.



Note: If a zero lies **on** the unit circle, the response becomes exactly zero at that zero's angle; if it lies inside but close to the unit circle, it produces a dip.

ELEC3104: Digital Signal Processing

Topic 3: Digital Filter Fundamentals

- FIR and IIR filters (Ref pp. 111-117)
- Digital Filter Structures (Ref pp. 334-343)
- Magnitude and Phase Responses (Ref pp. 308-352)
- Minimum, Maximum, Mixed Phase Filters (Ref PP. 362-368)
- All-Pass filters (Ref pp. 353-354)

Reference Book: J. Proakis & D. Manolakis, Digital Signal Processing, Prentice-Hall, 2014

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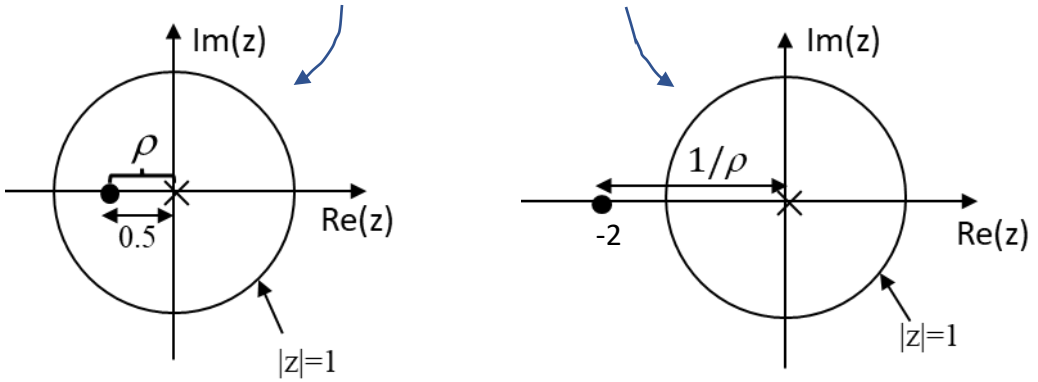
Term 3, 2026

Digital Filter Fundamentals: Minimum, Maximum & Mixed Phase Filters

FIR filter: Reciprocal roots

- The magnitude responses of two FIR filters $H_1(z)$ and $H_2(z)$ can be **identical** when their zeros are related by reciprocal reflection about the unit circle.
- If a zero inside the unit circle is reflected to its reciprocal-conjugate location outside the unit circle, the magnitude response remains unchanged, but the phase response changes.
- Let us consider two FIR filters of transfer functions :

$$H_1(z) = 1 + \frac{1}{2}z^{-1}; \qquad H_2(z) = \frac{1}{2} + z^{-1}$$



- $H_2(z)$ is the reverse of the system $H_1(z)$. This is due to the reciprocal relationship between the zeros of $H_1(z)$ and $H_2(z)$.

- Indeed, the coefficients are reversed : $H_1(z)$: [1, 1/2]; $H_2(z)$: [1/2, 1].
- $H_2(z)$ has the time-reversed coefficient sequence of $H_1(z)$, resulting in reciprocal zero locations.

Digital Filter Fundamentals: Minimum, Maximum & Mixed Phase Filters

$H_1(\theta) = 1 + (1/2)\, e^{-j\theta}$ $\quad$ $H_2(\theta) = 1/2 + e^{-j\theta}$

$$|H_1(\theta)| = |H_2(\theta)| = \sqrt{5/4 + \cos\theta}$$

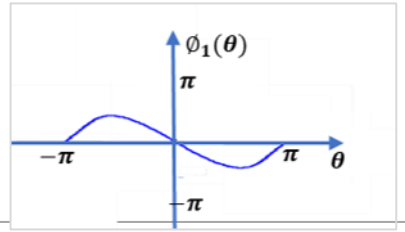
- Magnitude responses of $H_1(\theta)$ and $H_2(\theta)$ are identical
- **Phase response of $\boldsymbol{H_1(\theta)}$**

$$H_1(\theta) = \left(1 + \frac{1}{2}\cos\theta\right) - j\frac{1}{2}\sin\theta$$

Therefore, $\phi_1(\theta) = \tan^{-1}\left[\frac{-\frac{1}{2}\sin\theta}{1+\frac{1}{2}\cos\theta}\right]$

The boundary values:

$$\phi_1(-\pi) = 0, \phi_1(0) = 0, \phi_1(\pi) = 0.$$



- **Phase response of $\boldsymbol{H_2(\theta)}$:**

$$H_2(\theta) = \left(\frac{1}{2} + \cos\theta\right) - j\sin\theta$$

Therefore, $\phi_2(\theta) = \tan^{-1}\left[\frac{-\sin\theta}{\frac{1}{2}+\cos\theta}\right]$

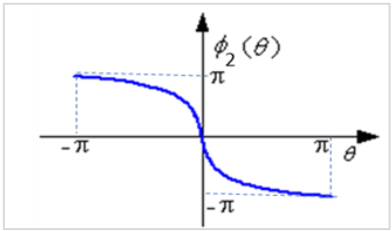
- The boundary values: $\phi_2(0) = 0.$

As $\theta \to \pi^-$, $\quad H_2(\theta) \to -\frac{1}{2} - j0,$

so $\quad \phi_2(\pi^-) \to -\pi$

- Similarly, as $\theta \to -\pi^+$, $\quad H_2(\theta) \to -\frac{1}{2} + j0,$

giving $\quad \phi_2(-\pi^+) \to +\pi$



Minimum-phase filter: All zeros lie inside the unit circle. Among filters with the same magnitude response, it has the minimum phase delay.

Maximum-phase filter: All zeros lie outside the unit circle. It has a larger phase delay than the corresponding minimum-phase filter with the same magnitude response.

Mixed-phase filter: Some zeros lie inside and some outside the unit circle. Its phase characteristic lies between those of the corresponding minimum- and maximum-phase forms.

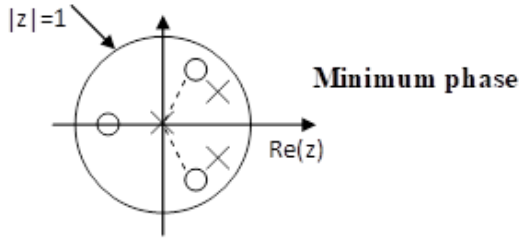
- Although $|H_1(\theta)| = |H_2(\theta)|$, their phases are very different.
- H_1, whose zero is at -0.5 **inside** the unit circle, has the smaller phase variation;
- H_2, whose zero is at -2 **outside** the unit circle, accumulates much greater phase shift.

Digital Filter Fundamentals: Minimum, Maximum & Mixed Phase Systems

Minimum, Maximum & Mixed Phase Filters

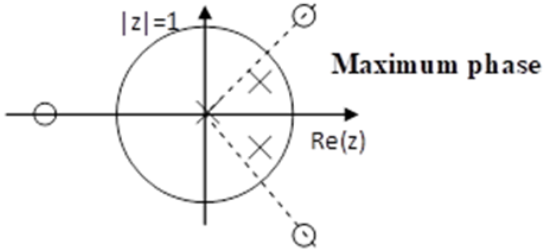
- The minimum-, maximum- and mixed-phase concepts for FIR filters can also be applied to IIR filters.
- Let us consider:

$$H(z) = \frac{B(z)}{A(z)}$$

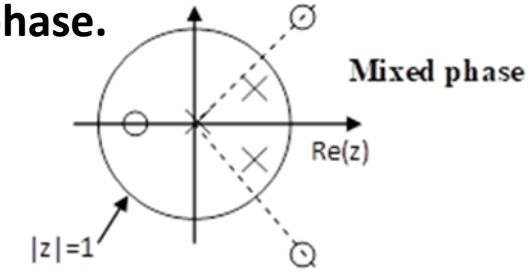


A causal, stable IIR system is called **minimum phase** if all its poles and zeros lie inside the unit circle.

- If all the zeros lie outside the unit circle, the system is called **maximum phase**.



- If some zeros lie both inside and outside the unit circle, the system is called **mixed-phase.**



Note:

- For a given magnitude response, the minimum-phase system has all its zeros inside the unit circle and has the smallest group delay among the corresponding causal, stable systems.
- Reflecting zeros from inside to outside the unit circle can preserve the magnitude response (with appropriate gain scaling), but changes the phase response and group delay.
- Therefore, systems can have the same magnitude response but different phase and delay characteristics, depending on the locations of their zeros.

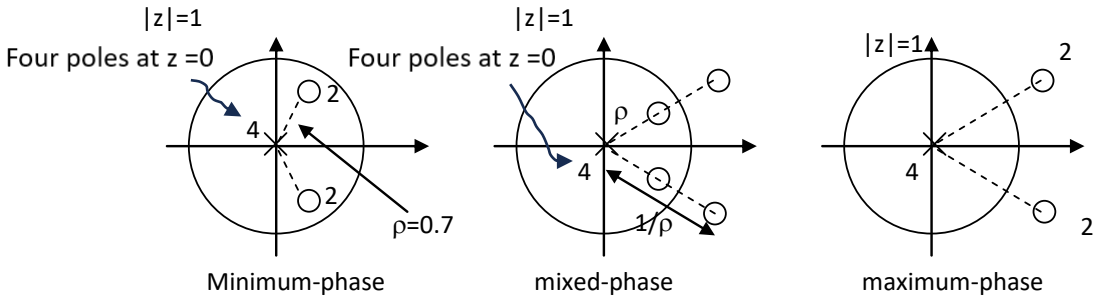
Digital Filter Fundamentals: Minimum, Maximum & Mixed Phase Systems

Example 20:

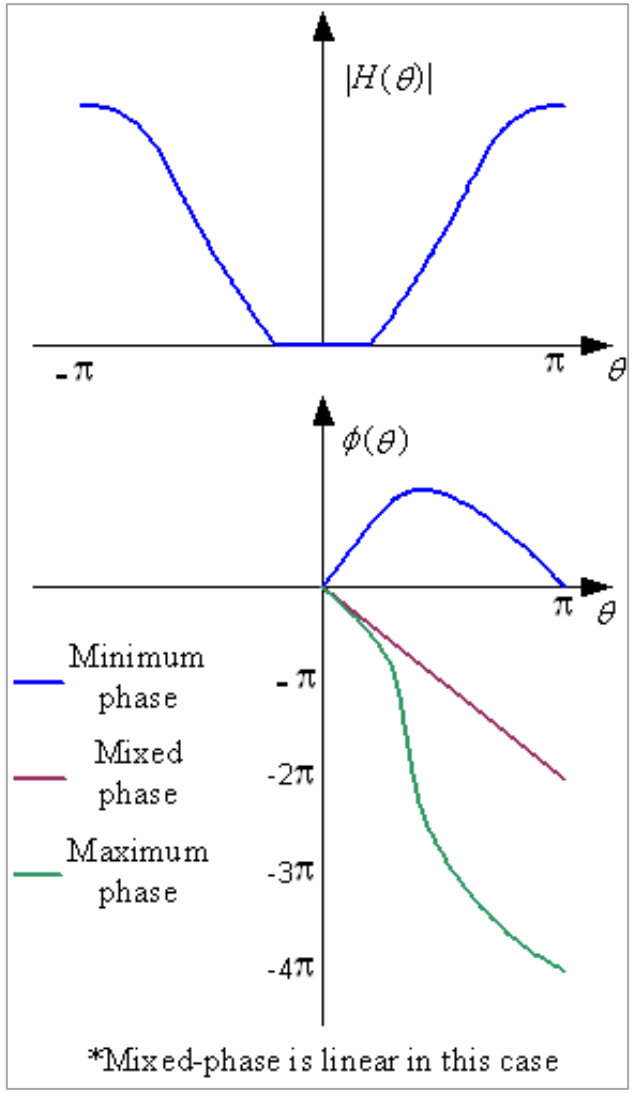
- Consider a fourth-order all-zero filter containing two identical complex conjugate pairs of zeros located at

$$z = 0.7e^{\pm j\frac{\pi}{4}}$$

- The The minimum-phase, mixed-phase, and maximum-phase systems obtained by reciprocal reflection of the zeros have identical magnitude responses (apart from a constant gain factor). Their pole-zero patterns are shown below.



- The magnitude response and the phase response of the three systems are shown on the right.
- Although all three systems have the same magnitude response, they have different phase responses. The minimum-phase system has the least overall phase accumulation, while the maximum-phase system has the greatest.



Four poles at z =0

Digital Filter Fundamentals: Minimum, Maximum & Mixed Phase Systems

Example 21:

- A third order FIR filter has a transfer function $G(z)$ given by:

$$G(z) = (6 - z^{-1} - 12z^{-2})(2 + 5z^{-1})$$

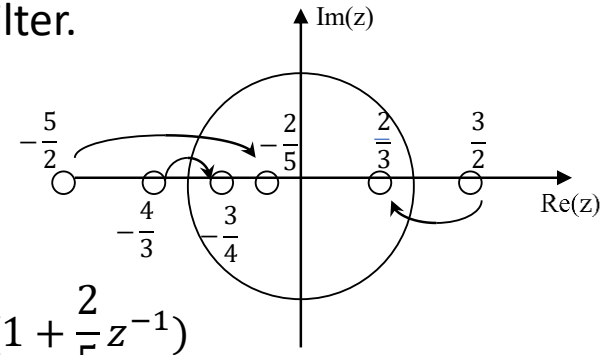
From $G(z)$, determine the transfer function of an FIR filter whose magnitude response is identical to that of $G(z)$ and has a minimum phase response.

Answer: First Factorise G(z):

$$G(z) = (2 - 3z^{-1})(3 + 4z^{-1})(2 + 5z^{-1})$$

$$G(z) = 12(1 - \frac{3}{2}z^{-1})(1 + \frac{4}{3}z^{-1})(1 + \frac{5}{2}z^{-1})$$

- All the zeros of G(z) lie outside the unit circle; Therefore reflect to its reciprocal location inside the unit circle to obtain the minimum-phase filter.

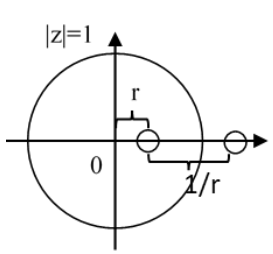


The minimum phase filter:

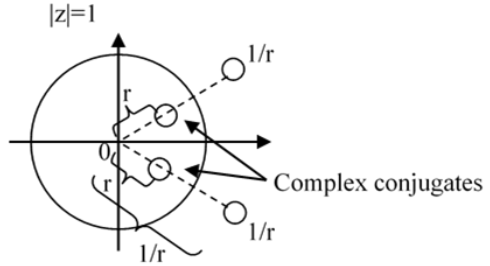
$$P(z) = 60(1 - \frac{2}{3}z^{-1})(1 + \frac{3}{4}z^{-1})(1 + \frac{2}{5}z^{-1})$$

Example 22: Reciprocal pairs of zeros for linear phase filters

- The zeros of a real-coefficient linear-phase FIR filter exhibit reciprocal and complex-conjugate symmetry as shown below.

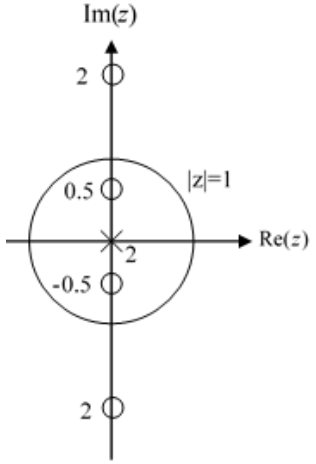


For a real zero



For a complex zero

- The pole-zero plot of a filter H(z) is given below. Is $H(z)$ a linear phase filter?



The four zeros at $z = j0.5$, $z = j2$, $z = -j0.5$ and $z = -j2$ are conjugate reciprocal symmetry. All poles are at $z = 0$.

Therefore, this is a linear phase filter.

$H(z) = z^2 + 4.25 + z^{-2}$

$\therefore h[n] = \{1, 0, 4.25, 0, 1\}$ <– even symmetry about n=0.

ELEC3104: Digital Signal Processing

Topic 3: Digital Filter Fundamentals

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- All-Pass filters (Ref pp. 353-354)

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Digital Filter Fundamentals: All-Pass filters

All-Pass Filters

- An all-pass filter is one whose magnitude response is constant for all frequencies, but whose phase response is not identically zero.
- The simplest example of an all-pass filter is a pure delay system with system function $H(z) = z^{-k} \quad k > 0$
- A more interesting all-pass filter is one that is described by

$$H(z) = \frac{a_L + a_{L-1}z^{-1} + \cdots + a_1 z^{-L+1} + a_0 z^{-L}}{1 + a_1 z^{-1} + \cdots + a_L z^{-L}}$$

where $a_0 = 1$ and all coefficients are real.

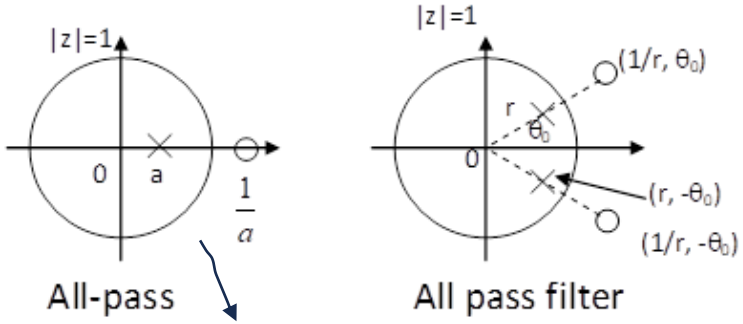
- If we define the polynomial $A(z)$ as

$$A(z) = \sum_{k=0}^{L} a_k z^{-k} \qquad a_0 = 1$$

$$\therefore H(z) = z^{-L}\frac{A(z^{-1})}{A(z)}$$

- It can be shown that $|H(\theta)|^2 = H(z)\cdot H(z^{-1})|_{z=e^{j\theta}} = 1$ which is an all-pass filter.

- Furthermore, if $|z_0|$ is the modulus of a pole of $H(z)$, then $1/|z_0|$ is the modulus of a zero of $H(z)$ {i.e. the modulus of poles and zeros are reciprocals of one another}.
- The figure shown below illustrates typical pole-zero patterns for a single-pole, single-zero filter and a two-pole, two-zero filter.



$$H(z) = \frac{1-\frac{1}{a}z^{-1}}{1-az^{-1}} \qquad |a| < 1 \text{ for stability}$$

- We can easily show that the magnitude response is a constant.

$$|H(\theta)|^2 = |H(\theta)\cdot H^*(\theta) = H(z)\cdot H(z^{-1})|_{z=e^{-j\theta}}$$

$$= \frac{1-\frac{1}{a}e^{-j\theta}}{1-ae^{-j\theta}}\cdot\frac{1-\frac{1}{a}e^{j\theta}}{1-ae^{j\theta}} = \frac{1-\frac{2}{a}\cos\theta+\frac{1}{a^2}}{1-2a\cos\theta+a^2} = 1/a^2$$

Digital Filter Fundamentals: All-Pass filters

Example 23:

- Show that the following transfer function $H_1(z)$ can be obtained using a parallel connection of a constant and an all-pass filter.

$$H_1(z) = \frac{10 + 6z^{-1}}{3 + z^{-1}}$$

Answer:

$$H_1(z) = \frac{9 + 3z^{-1} + 1 + 3z^{-1}}{3 + z^{-1}}$$

$$= 3 + \frac{1 + 3z^{-1}}{3 + z^{-1}}$$

$$H_1(z) = \underbrace{3}_{constant} + \underbrace{\frac{1 + 3z^{-1}}{3 + z^{-1}}}_{All-passfilter}$$

Example 24: All-pass filter

- Consider the filter described by

$H(z) = \frac{a+cz^{-2}}{c+az^{-2}}$ where a & c are constants.

Show that the magnitude response $|H(\theta)|$ is unity for all θ.

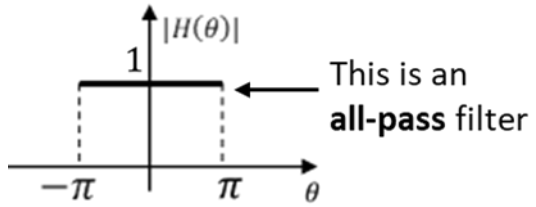
Answer:

$$H(z) = \frac{c+az^{-2}}{a+cz^{-2}}; \qquad H(\theta) = \frac{c+ae^{-j2\theta}}{a+ce^{-j2\theta}}$$

$$|H(\theta)|^2 = H(\theta) \cdot H^*(\theta)$$

$$= \frac{c + ae^{-j2\theta}}{a + ce^{-j2\theta}} \cdot \frac{c + ae^{j2\theta}}{a + ce^{j2\theta}}$$

$$= \frac{c^2 + a^2 + ac[e^{j2\theta} + e^{-j2\theta}]}{a^2 + c^2 + ac[e^{j2\theta} + e^{-j2\theta}]} = 1$$



Digital Filter Fundamentals: All-Pass filters

Example 25:

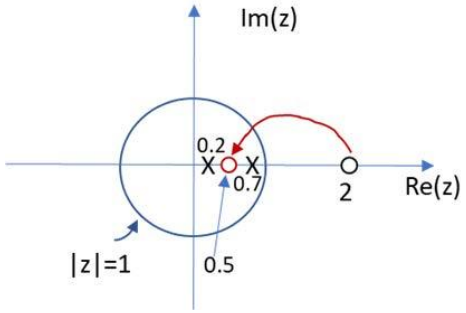
- A stable and causal non minimum phase filter may always be factored into a product of a minimum phase filter (with all poles and zeros inside the unit circle) with an all-pass filter.

$$H(z) = H_{min}(z).\, H_{allpass}(z).$$

- The procedure for performing this factorisation is as follows:
  - ❖ First, all of the zeros of H(z) that are outside the unit circle are reflected inside the unit circle to their conjugate reciprocal location. The resulting filter is minimum phase, $H_{min}(z)$.
  - ❖ Then, all-pass filter is selected so that it reflects the appropriate set of zeros of $H_{min}(z)$, back outside the unit circle.

- For the transfer function $H(Z) = \frac{1-2z^{-1}}{(1-0.2\,z^{-1})(1-0.7z^{-1})}$, find the factorisation for H(z) of the form

$$H(z) = H_{min}(z).\, H_{allpass}(z).$$



- The minimum phase factor is: $H_{min}(z) = \frac{1-\left(\frac{1}{2}\right)z^{-1}}{(1-0.2\,z^{-1})(1-0.7z^{-1})}$
- Then , to reflect the zero at $z = 0.5$ back outside the unit circle to $z = 2$, we use the all-pass filter.
- $H_{Allpass}(z) = \frac{1-2\,z^{-1}}{1-\left(\frac{1}{2}\right)z^{-1}} \leftarrow All\ Pass\ filter$

$$H(Z) = \frac{1-\left(\frac{1}{2}\right)z^{-1}}{(1-0.2\,z^{-1})(1-0.7z^{-1})} \cdot \frac{1-2\,z^{-1}}{1-\left(\frac{1}{2}\right)z^{-1}}$$

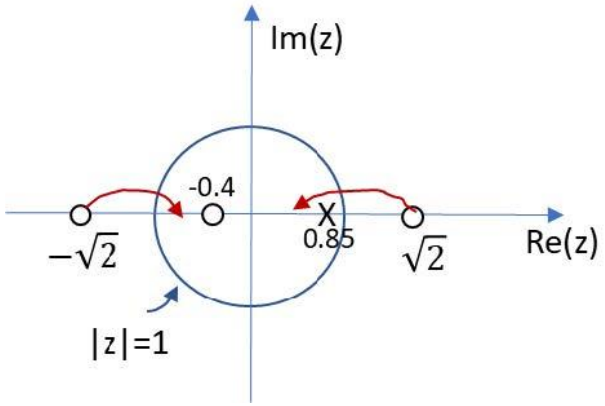
$$H(z) = \quad H_{min}(z). \quad H_{allpass}(z).$$

Digital Filter Fundamentals: All-Pass filters

Example 26:

- A causal LTI system has a transfer function: $H(Z) = \frac{(1-2z^{-2})\ (1+0.4z^{-1})}{(1-0.85\ z^{-1})}$; Find a factorisation for H(z) of the form $H(z) = H_{min}(z).\ H_{allpass}(z)$, where $H_{min}(z)$ is a minimum phase system and $H_{allpass}(z)$ is a stable, all-pass system.

The pole-zero map of H(z) is given by:



The system function H(z) has a nonminimum phase factor, $(1 - 2z^{-2})$, which may be written as the product of a minimum phase term and an all-pass factor as follows:

$$(1 - 2z^{-2}) = (1-\left(\frac{1}{2}\right) z^{-2}) \cdot \frac{(1-2z^{-2})}{\left(1-\left(\frac{1}{2}\right)z^{-2}\right)}$$

Therefore , H(z) may be written as the product of a minimum phase system with an all-pass system as follows:

$$H(Z) = \frac{(1+0.4z^{-1})(1-\left(\frac{1}{2}\right)z^{-2})}{(1-0.85\ z^{-1})} \qquad \frac{(1-2z^{-2})}{\left(1-\left(\frac{1}{2}\right)z^{-2}\right)}$$

$H_{min}(z)$ $H_{allpass}(z)$

Topic 3- Summary

Summary

1. Types of digital filters (**FIR and IIR**) and their properties.
2. Conversion between **difference equations and transfer functions.**
3. **FIR filters:** non-recursive and all-zero structures.
4. **Phase delay and group delay**, and their interpretation.
5. **Linear-phase FIR filters** and the relationship between coefficient symmetry and linear phase.
6. **IIR filters**: recursive, all-pole and pole-zero structures.
7. **Digital filter structures:** Direct Form I, Direct Form II, canonical, cascade and parallel structures.
8. Calculation of **DC gain, 3-dB cut-off frequency** and 3-dB bandwidth for simple FIR and IIR filters.
9. Sketching **magnitude and phase responses** of simple FIR and IIR filters.
10. Identifying **low-pass and high-pass filters** from their difference equations, transfer functions and pole-zero locations.
11. Understanding the relationship between pole-zero locations and the magnitude and phase responses.
12. **Minimum-, maximum- and mixed-phase filters**, and their zero locations relative to the unit circle.
13. Understanding how reciprocal reflection of zeros about the unit circle affects magnitude and phase responses.
14. **All-pass filters:** showing that the magnitude response is constant while the phase response is frequency dependent.
15. Decomposition of a non-minimum-phase system into a minimum-phase system and an all-pass system